



Review article

Decoding the secretase puzzle in amyloid- β generation: A state-of-the-art overview of the protease-mediated APP processing cascade in Alzheimer's disease

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ABSTRACT

The accumulation of amyloid β (A β) protein in the brain is a central pathological hallmark of Alzheimer's disease (AD). This process has become a major focus of interdisciplinary research and a critical target in drug development. A β is produced through the proteolytic processing of amyloid precursor protein (APP) by a group of enzymes known as secretases. They belong to different protease classes and operate through proteolytic cleavage of the peptide bond through several catalytic hydrolysis. Dysregulation of the expression or/and activity of proteases involved in APP processing disrupts the balance between the amyloidogenic and non-amyloidogenic pathways—often shifting it toward the amyloidogenic route. This shift leads to excessive production and further aggregation of A β peptides, ultimately resulting in neuronal toxicity. In this review, we integrate current state-of-the-art knowledge on all proteases reported to cleave APP, encompassing both canonical and non-canonical pathways, and offer detailed examination of cleavage-site topology, and catalytic mechanisms. By integrating the spatial and sequential hierarchy of APP proteolysis across cellular compartments, we establish a unifying mechanistic framework that captures the complexity of the process. We further delineate how distinct proteases—through defined active-site architectures, conserved catalytic motifs, and nucleophile-driven peptide bond hydrolysis—precisely regulate APP processing. This mechanistic perspective advances our molecular understanding of A pathogenesis and delineates critical catalytic control nodes amenable to therapeutic intervention. By defining these targets at a mechanistic level, it establishes a rational framework for precision drug design and the development of next-generation therapeutics.

1. Introduction

Society is undergoing demographic shifts, marked by longer life expectancy and lower birth rates, leading to an aging population. As a result, age-related diseases are becoming more prevalent (Atella et al., 2019; Jaul and Barron, 2017). One such condition is dementia, a group of symptoms that includes impairments in memory, language, problem-solving, and other cognitive functions, all of which disrupt a person's ability to perform daily activities (Alzheimer's Association, 2024). Alzheimer's disease (AD), along with Parkinson's disease (PD), Lewy body disease, hippocampal sclerosis, and cerebrovascular disease, is a major cause of dementia (Alzheimer's Association, 2024; Ball et al., 2020), accounting for 60–80% of all cases. In 2024, approximately 6.9 million Americans aged 65 and older are living with Alzheimer's

dementia. By 2060, this number is expected to rise to 13.8 million, assuming no medical breakthroughs are made to prevent or cure this disorder. Data from the Centers for Disease Control and Prevention shows that 119,399 people died from AD in 2021 (Alzheimer's Association, 2024; Niu et al., 2017; Roberts et al., 2024; Saraceno et al., 2024; Tomaskova et al., 2016).

The most well-known condition linked to amyloid β (A β) pathology is AD, where A β peptides aggregate into oligomers and fibrils, leading to synaptic dysfunction, neuroinflammation, and neuronal death (Fig. 1A and B). Among the various forms of A β , A β 42 is considered the most toxic and plays a central role in plaque formation (Abedin et al., 2021; Ono and Watanabe-Nakayama, 2021; Jaul and Barron, 2017). Another condition heavily associated with A β accumulation is Cerebral Amyloid Angiopathy (CAA), in which A β deposits in the walls of cerebral blood

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vessels (Cozza et al., 2023), increasing the risk of microbleeds, strokes, and cognitive impairment. CAA often coexists with AD but can also occur independently (Kozberg et al., 2021). Individuals with Down syndrome (DS; Trisomy 21) are also prone to A β accumulation due to an extra copy of the amyloid β precursor protein (APP) gene. As a result, nearly all individuals with DS develop Alzheimer's-like pathology, including A β plaques, by middle age (Fernandez et al., 2024; Krasny et al., 2024; Tosh et al., 2021). Other neurodegenerative diseases, such as Parkinson's Disease with Dementia (PDD) (Lim et al., 2019; Mihaescu et al., 2022) and Dementia with Lewy Bodies (DLB) (van Steenoven et al., 2019), also show A β pathology. While these conditions are primarily associated with α -synuclein aggregates, A β plaques are often present (Han et al., 2021), especially in DLB (Biundo et al., 2021), and may contribute to cognitive decline (Jellinger and Korszyn, 2018). Similarly, in Huntington's disease, though the primary pathology is driven by mutant huntingtin protein (Farag et al., 2024), A β deposits have been observed in some cases and may play a role in disease progression (Korpela et al., 2024; Zhang et al., 2016). Multiple system atrophy, a rare neurodegenerative disorder mainly linked to α -synuclein accumulation (Schweighauser et al., 2020; Shahnawaz et al., 2020), has also been found to exhibit A β plaques in certain cases (Kon et al., 2024). Furthermore, brain injuries can trigger A β deposition, as seen in both traumatic brain injury (Özen et al., 2025) and chronic traumatic encephalopathy (Ramos-Cejudo et al., 2018; Stein et al., 2015). Repeated head trauma, common in athletes and military personnel, has been linked to early A β plaque formation, contributing to progressive neurodegeneration similar to AD (Lee et al., 2019; McKee et al., 2016). Thus, A β accumulation is not exclusive to AD; it is a shared pathological feature observed in several neurodegenerative and brain-related disorders, offering a broader context for its role in disease and therapeutic targeting (Fig. 1A).

In AD specifically, the pathological aggregation of A β and tau proteins—forming extracellular plaques and intracellular neurofibrillary tangles (NFTs), respectively—constitutes two defining hallmarks of the disease (Tan et al., 2014; Zhang et al., 2022). While their accumulation

is closely linked to selective neuronal loss and widespread brain damage, the pathophysiology of AD extends beyond these hallmarks (Edler et al., 2017; Hampel et al., 2021). It encompasses a multifaceted interplay of contributing factors, including genetic predisposition (Bellenguez et al., 2022), oxidative stress (Dhapola et al., 2024), neuroinflammation (Heneka et al., 2024), and impaired cholinergic neurotransmission (Akyuz et al., 2025). Additionally, environmental factors such as air pollution, diets rich in saturated fats and calories, and exposure to toxic metals further exacerbate disease progression (Breijyeh and Karaman, 2020; Rahman et al., 2020; Zhao et al., 2021) (Fig. 1C). Understanding the involvement of A β across various conditions and its integration with these complex pathological mechanisms is essential for developing more effective and comprehensive therapeutic strategies.

Proteases play a central role in the fragmentation of amyloid precursor protein (APP), as their precise cleavage determines whether APP is processed through the non-amyloidogenic or amyloidogenic pathway. By controlling the generation of A β peptides and other bioactive fragments, proteases directly influence the molecular events underlying Alzheimer's disease pathology. This review is designed to provide a comprehensive and integrated understanding of APP proteolytic cleavage, with a particular focus on cataloging all secretases involved across all known processing pathways based on the current state of the art. By combining a precise analysis of APP processing, subcellular localization, cleavage sites with insights from medicinal chemistry and enzymatic catalysis perspective, this work aims to clarify the molecular basis of A β generation—a central event in the pathogenesis of AD. We detail the roles of secretases in both amyloidogenic and non-amyloidogenic pathways, shedding light on their substrate specificity and mechanistic nuances.

Uniquely, this review bridges neurobiological and chemical perspectives, offering both neuroscientists and drug designers a coherent framework for understanding how these proteases function and how they might be modulated therapeutically. In doing so, we aim to support the development of more effective, targeted interventions for AD and related neurodegenerative disorders.

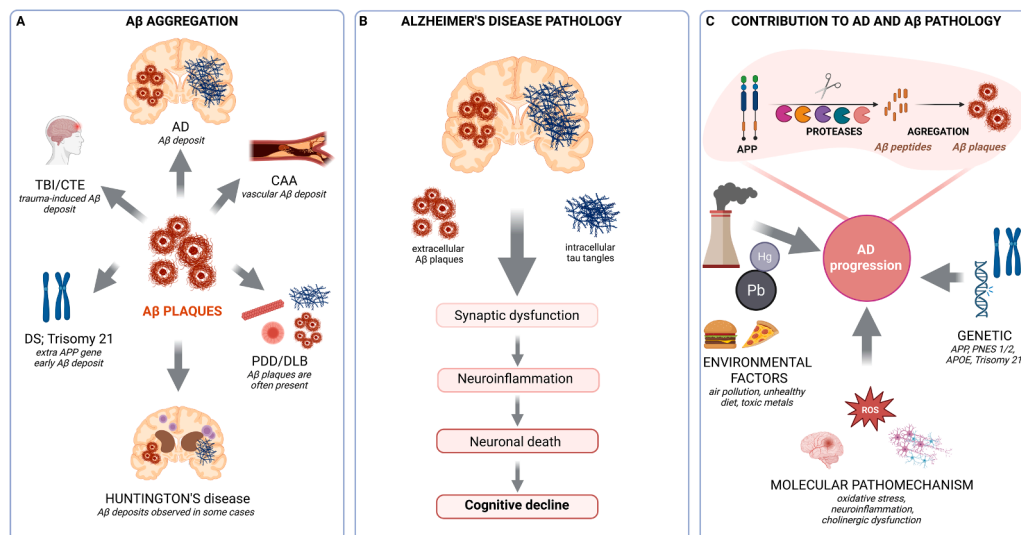


Fig. 1. Amyloid- β (A β) pathology across neurological disorders and its central role in Alzheimer's disease (AD) progression. (A) Schematic representation of A β plaque formation and its association with multiple neurological conditions including Alzheimer's disease (AD), cerebral amyloid angiopathy (CAA; vascular A β deposition), traumatic brain injury and chronic traumatic encephalopathy (TBI and CTE), Down syndrome (Trisomy 21; extra APP gene copy leading to early A β accumulation), and other neurodegenerative conditions such as Parkinson's disease with dementia (PDD), dementia with Lewy bodies (DLB), multiple system atrophy (MSA), and Huntington's disease. (B) Core pathological cascade in Alzheimer's disease. Extracellular A β plaques and intracellular tau neurofibrillary tangles (NFTs) represent the defining hallmarks of AD. Their accumulation leads to synaptic dysfunction, activation of neuroinflammatory responses, progressive neuronal death, and ultimately cognitive decline. (C) Multifactorial contributors to AD progression. Proteolytic processing of amyloid precursor protein (APP) generates A β peptides that initiate pathological aggregation. AD progression is influenced by genetic predisposition (e.g., APP, PSEN1/2, APOE), oxidative stress (reactive oxygen species, ROS), neuroinflammation, environmental risk factors (air pollution, unhealthy diet, toxic metals such as Hg and Pb), and other systemic contributors. Created with BioRender.com.

2. Pathomechanism of AD

2.1. Familial (fAD/EOAD) vs sporadic (sAD/LOAD)

Alzheimer's disease (AD) is a clinically and genetically heterogeneous neurodegenerative disorder that is broadly classified into two principal forms: familial AD (fAD) and sporadic AD (sAD). Familial AD, also referred to as early-onset AD (EOAD), accounts for approximately 1–6% of all cases and typically manifests before the age of 65 (Ulaganathan, Pitchaimani, 2023). It follows an autosomal dominant pattern of inheritance driven by highly penetrant mutations in three genes: the amyloid precursor protein gene (APP) on chromosome 21, and the presenilin genes PSEN1 and PSEN2, which encode the catalytic subunits of the γ -secretase complex (Cacace et al., 2016; Bagaria et al., 2022; Tcw and Goate, 2017). Mutations in these genes converge on a single downstream effect: increased production of the aggregation-prone A β 42 isoform and/or an elevated A β 42/A β 40 ratio, directly accelerating amyloid plaque formation and disease onset (Müller et al., 2017). PSEN1 mutations are the most prevalent, accounting for 18–50% of fAD cases, while PSEN2 and APP mutations are less frequent but equally deterministic in their pathogenic consequence (Alonso Vilatela et al., 2012; Jiang et al., 2018). In contrast, sporadic AD (sAD), also known as late-onset AD (LOAD), constitutes more than 90% of all AD cases and typically manifests after the age of 65 (Piaceri et al., 2013). Unlike fAD, sAD does not arise from single, highly penetrant mutations but rather from a complex, multifactorial interplay of genetic susceptibility loci, aging-related biological changes, and environmental exposures.

A key mechanistic distinction between the two forms lies in the primary driver of A β pathology: fAD is characterised by overproduction of A β owing to structural changes in APP processing enzymes, whereas sAD is predominantly driven by age-related impairment of A β clearance mechanisms—including reduced activity of neprilysin and insulin-degrading enzyme, decreased LRP1-mediated efflux across the blood–brain barrier, and glymphatic dysfunction—combined with increased activity of the proteases involved in APP processing (Tarasoff-Conway et al., 2015; Pera et al., 2013).

Among the key contributors to Alzheimer's disease pathogenesis are oxidative stress and mitochondrial dysfunction (Veselov et al., 2023), cholinergic alterations (Campos-Peña et al., 2022), genetic factors (Zheng and Wang, 2025), neuroinflammation (Twarowski and Herbet, 2023), microRNA dysregulation (Dalal et al., 2024; Sharma et al., 2024), a synaptic damage (Cheng et al., 2025), gut microbiome alterations (Jiang et al., 2017), and impaired autophagy (Uddin et al., 2019). These processes are observed in both familial and sporadic forms of AD; however, while they primarily represent downstream consequences of A β overproduction in familial AD, in sporadic AD they act as key drivers that both promote amyloidogenic APP processing and amplify disease progression.

The strongest genetic risk factor for sAD is the ϵ 4 allele of the apolipoprotein E gene (APOE ϵ 4), which increases AD risk approximately 3–4-fold in heterozygous and 8–12-fold in homozygous carriers compared to the common ϵ 3 allele (Liu et al., 2013; Raulin et al., 2022). ApoE4 impairs A β clearance, promotes its aggregation, and disrupts lipid metabolism and endosomal trafficking, thereby amplifying amyloidogenic processing without directly altering APP or secretase gene structure (Lista et al., 2015).

Additional genome-wide association studies have implicated numerous low-to-moderate risk loci in sAD, including CLU, BIN1, CR1, and TREM2, which collectively modulate immune surveillance, endocytosis, lipid transport, and complement activity in ways that collectively lower the threshold for A β accumulation and neuroinflammation (Aerqin et al., 2022; Zheng and Wang, 2025). Despite their distinct aetiological origins, both fAD and sAD converge on the same core neuropathological features: extracellular deposition of A β plaques, intracellular accumulation of neurofibrillary tangles (NFTs) composed

of hyperphosphorylated tau, progressive synaptic loss, neuroinflammation, and ultimately widespread neurodegeneration.

Although these forms share many pathological features and overlapping molecular pathways, including mechanisms involving protease-mediated APP processing—the central focus of this review—they differ in their primary initiating events and relative contributions of these pathways to disease onset and progression.

This mechanistic convergence justifies a unified pathomechanistic framework while recognising that targeted therapeutic strategies may need to be tailored to the predominant pathogenic driver in each form—A β overproduction in fAD versus impaired clearance and broader cellular dysfunction in sAD (Villain et al., 2022; Almeida et al., 2020).

2.2. Proteolytic processing of APP and secretase biology

A hallmark pathological feature of Alzheimer's is the accumulation of amyloid β peptides, which are generated through the proteolytic processing of APP (Müller et al., 2017). The production of A β stems from the enzymatic cleavage of APP by secretases (Fig. 2 A), enzymes that belong to various protease families and function via distinct catalytic mechanisms to break peptide bonds (Landman and Kim, 2004). When the regulation of these proteolytic enzymes is disrupted, APP is increasingly processed through the amyloidogenic pathway. This shift results in the overproduction and accumulation of A β , promoting its aggregation and triggering neurotoxic effects. Understanding the intricate regulation of proteolytic events in the process of A β formation provides valuable insights into the development of AD and many other diseases (Zhang et al., 2012; Żukowska et al., 2024).

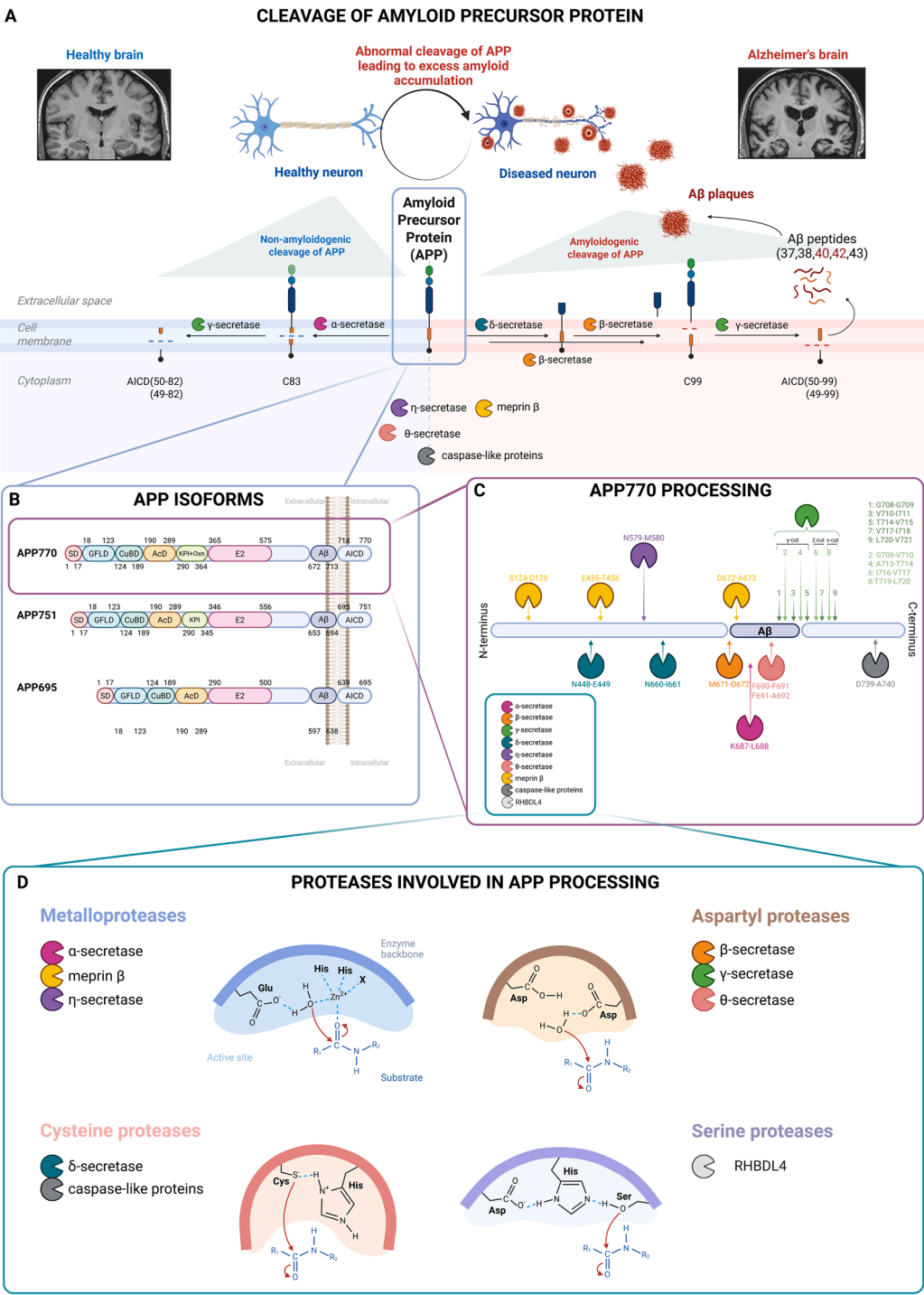
The differential processing of APP in healthy and Alzheimer's disease conditions is illustrated in Fig. 2 A. In the healthy brain, APP undergoes predominantly non-amyloidogenic cleavage mediated by α -secretase within the A β sequence, precluding A β formation. Subsequent γ -secretase cleavage generates soluble fragments and the intracellular domain AICD (50–82/49–82), along with the membrane-bound fragment C83 (Azargoonjahromi, 2024). This pathway prevents amyloid plaque formation and is associated with normal neuronal structure and brain morphology. In contrast, in Alzheimer's disease, APP is processed via the amyloidogenic pathway (Fig. 2 A and 2 C). Initial cleavage by β -secretase produces the C99 fragment, followed by γ -secretase cleavage, generating A β peptides of varying lengths (e.g., A β 37, A β 38, A β 40, A β 42) (Bhatia et al., 2023). These peptides aggregate extracellularly to form A β plaques, contributing to neuronal dysfunction and brain atrophy. Alternatively, δ -secretase cleaves APP to generate a fragment that is more susceptible to subsequent β -secretase cleavage (Meng et al., 2025; English et al., 2021). Additional proteases, including η -secretase, θ -secretase, meprin β , and caspase-like proteins, are shown as alternative cleavage contributors (Marengo et al., 2022; Sáez-Valero and Pérez-González, 2023; Xu et al., 2024).

APP belongs to the type I single-pass transmembrane protein family with a short cytoplasmic domain (C-terminus) and a long extracellular domain (N-terminus) (Hampel et al., 2021) (Fig. 2B). It is widely expressed by brain neurons, vascular endothelial cells, blood cells (including platelets), as well as, to a lesser extent, astrocytes (Hampel et al., 2021). APP is encoded by a single gene located on chromosome 21 in the human genome (Nguyen, 2019). This gene is composed of 18 exons which through alternative splicing may produce three primary isoforms: APP695, APP751, and APP770 (Fig. 1B). Human APP770 contains several functional segments within its ectodomain: the cysteine-rich E1 domain (consists of growth factor-like domain (GFLD) and copper-binding domain (CuBD)), an acidic domain (AcD), the Kunitz protease inhibitor (KPI) domain, the Ox-2 domain, and the helix-rich E2 domain (consists of RERMS sequence and a central APP domain (CAPPD)). This ectodomain also includes part of the A β sequence, which continues into the transmembrane region. Adjacent to A β lies the APP intracellular domain (AICD), produced when γ -secretase cleaves the C-terminal fragment, spanning both transmembrane and

cytoplasmic regions. Alternative isoforms differ in composition: APP751 lacks the Ox-2 domain, while APP695 is missing both the KPI and Ox-2 domains. The extracellular domains (E1 and E2) as well as the cytoplasmic tail are conserved across species. Isoform expression patterns are tissue-specific. APP695 is the principal isoform in the brain, whereas APP770 and APP751 predominate in astrocytes and peripheral tissues (Tiwari et al., 2019; Cho et al., 2022; Dai et al., 2017).

Full-length APP is synthesized in the endoplasmic reticulum (ER) and transported to the cell surface via the Golgi/trans-Golgi network (Jiang et al., 2014). During transport, APP undergoes multiple

post-translational modifications, including N- and O-linked glycosylation, phosphorylation, ubiquitination, and tyrosine sulfation (Ramazi et al., 2024). Numerous pathogenic mutations in APP associated with early-onset AD (EOAD) have been identified, along with a rare protective mutation (A673T) (Cacace et al., 2016; Jonsson et al., 2012). These pathogenic mutations are located near the β -secretase cleavage site (amino acids 670–682), the γ -secretase cleavage site (amino acids 713–724), or within the A β sequence (amino acids 692–705) of the protein (Dai et al., 2017), resulting in changes in efficacy of processing by secretases, which leads to an increased overall production of A β



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Fig. 2. (A) Schematic overview of amyloid precursor protein (APP) cleavage pathways in healthy and Alzheimer's disease (AD) brains. In the non-amyloidogenic pathway, α - and γ -secretase processing prevents A β formation and generates C83 and AICD fragments. In the amyloidogenic pathway, β -, δ - and γ -secretase cleavage produces A β peptides (e.g., A β 37–43), which aggregate extracellularly into amyloid plaques, contributing to neuronal dysfunction and brain atrophy. Alternative proteases are also indicated. The diagram highlights the extracellular space, cell membrane, and cytoplasmic compartments, as well as the positions of cleavage events and resulting fragments, emphasizing the molecular mechanisms underlying amyloid accumulation in AD. (B) Schematic representation of the full-length APP 770, 751, 695 isoforms domains. SD (Signal Domain) – residues 1–17, present in all isoforms; GFLD (Growth Factor-like Domain) – residues 18–123; CuBD (Copper Binding Domain) – residues 124–189; AcD (Acidic Domain) – residues 190–289; KPI (Kunitz Protease Inhibitor domain) – present only in APP770 and APP751, absent in APP695; Oxn (Oxidative stress-related domain) – part of KPI+Oxn in APP770; E2 domain – spans residues 290–575 (slightly shorter in APP751 and APP695); A β domain (Amyloid- β) – residues 672–713 (APP770), 653–694 (APP751), 597–638 (APP695); AICD (APP Intracellular Domain) – short cytoplasmic tail following the transmembrane region. (C) An overview of proteolytic processing of amyloid precursor protein 770 (APP770) by proteases. Enzyme are indicated by color as follows: **burgundy**, α -secretase activity mediated primarily by ADAM10 metalloprotease; **orange**, β -secretase cleavage mediated by the aspartyl protease BACE1; **green**, γ -secretase cleavage mediated by the presenilin-containing γ -secretase complex (presenilin-1/2, nicastrin, Aph-1, and Pen-2), including ϵ -, ζ -, and γ -cuts within the transmembrane domain that generate A β peptides of varying lengths; **turquoise**, δ -secretase cleavage mediated by the asparagine endopeptidase (AEP); **purple**, η -secretase, mediated by membrane-type 5-matrix metalloproteinase; **pink**, θ -secretase mediated by BACE2; **yellow**, meprin β -mediated cleavage; **gray**, cathepsin-mediated cleavage; **light gray**, endoplasmic reticulum-resident intramembrane rhomboid protease (RHBDL4) mediated cleavage. The A β sequence is highlighted, and the amino acid positions corresponding to each cleavage event are indicated, except in the case of RHBDL4. (D) Schematic representation of the four major protease classes involved in amyloid precursor protein (APP) processing in Alzheimer's disease distinguished by the nature of their active-site catalytic residues and mechanisms of peptide bond hydrolysis. **Metalloproteases** (top left) utilize a Zn²⁺ ion coordinated by histidine residues and a glutamate to activate a water molecule, which acts as the nucleophile for peptide bond cleavage. This class includes α -secretase, meprin β , and η -secretase. **Aspartyl proteases** (top right) contain two catalytic aspartate residues that activate a water molecule via general acid–base catalysis, leading to hydrolysis of the peptide bond. Members include β -secretase (BACE), γ -secretase, θ -secretase, and caspase-3. **Cysteine proteases** (bottom left) employ a catalytic cysteine–histidine dyad, in which the cysteine thiol serves as the nucleophile attacking the peptide bond carbonyl carbon. δ -secretase is shown as a representative enzyme. **Serine proteases** (bottom right) use a classical catalytic triad composed of serine, histidine, and aspartate residues, where the serine hydroxyl group performs a nucleophilic attack on the substrate. RHBDL4 is illustrated as an example. In all panels, the enzyme backbone is depicted schematically, key catalytic residues are highlighted, and the substrate peptide (R₁–CO–NH–R₂) is shown undergoing cleavage. Red arrows indicate the direction of nucleophilic attack and peptide bond hydrolysis. Created with BioRender.com.

and/or altering the ratio of specific A β peptides (Müller et al., 2017). There are several pathways that result in the production of A β (Tiwari et al., 2019), all of which depend on the sequential proteolytic processing of amyloid precursor protein (APP) (Fig. 2 C). These cleavage events are mediated primarily by secretases—specialized proteases that determine whether APP undergoes non-amyloidogenic or amyloidogenic processing. β -secretase (BACE1) initiates the amyloidogenic pathway by cleaving APP at the N-terminus of the A β sequence, followed by intramembrane cleavage by the γ -secretase complex, which releases A β peptides of varying lengths. Secretases belong to a broader and structurally diverse family of proteases, ranging from relatively small single-polypeptide enzymes (~20 kDa) to large, multi-subunit proteolytic assemblies such as the γ -secretase complex, which can reach molecular masses of 0.7–6 MDa. These enzymes degrade target proteins or peptides by cleaving α -peptide and isopeptide bonds, or by self-hydrolyzing their own polypeptide chains (Rai et al., 2022; Zmudzinski et al., 2023) (Fig. 2D). Proteases were initially divided into two main categories: endopeptidases, which cleave internal peptide bonds (Muñoz Solano et al., 2012), and exopeptidases, including aminopeptidases and carboxypeptidases, which act on the N-terminal (NH₂) and C-terminal (COOH) ends of substrates. However, with advances in structural and mechanistic understanding, a more refined classification emerged (López-Otín and Bond, 2008). Proteases are now categorized into six major classes based on their catalytic mechanisms: serine, cysteine, threonine, aspartic, metalloproteases, and glutamic proteases (Botos and Wlodawer, 2007). However, only four of these classes—metalloproteases, aspartic proteases, cysteine proteases, and serine proteases—have been implicated in amyloid precursor protein (APP) processing (Fig. 2D). These classes encompass the principal secretases and related enzymes responsible for both amyloidogenic and non-amyloidogenic cleavage pathways, thereby directly influencing A β generation and its pathological consequences. Notably, glutamic proteases have not yet been identified in mammals (Sarode et al., 2024). The first three classes use specific amino acid residues (Cys, Ser, or Thr) in the active site as nucleophiles, which is reflected in their respective names, while the latter classes employ an activated by ion water molecule as a nucleophile to cleave peptide bonds (Jeyaraju et al., 2023; López-Otín and Bond, 2008). The regulation of protease activity is crucial; while some proteases contribute to neurodegeneration by excessively cleaving proteins, others may provide protective effects against AD pathology by aiding in the degradation of toxic proteins

(Ramazi et al., 2024).

Once generated as soluble monomers, A β exists in various intermediate aggregation states, such as dimers, trimers, soluble oligomers, and protofibrils, before ultimately forming fibrils that accumulate in extracellular plaques, commonly recognized as a neuropathological hallmark of AD (Hampel et al., 2021).

2.3. Interplay between amyloid- β and other pathogenic processes in Alzheimer's disease

In sporadic Alzheimer's disease, neurodegeneration arises from a dynamic interplay between A β and multiple contributing factors that, although also present in familial AD, play a more central role in the sporadic form (Fig. 3). While these processes occur in both forms of the disease, they primarily represent downstream consequences of A β overproduction in familial AD, whereas in sporadic AD they function as key drivers that promote amyloidogenic APP processing and accelerate disease progression.

A β toxicity is largely mediated through disruption of calcium homeostasis and membrane integrity, caused by lipid disturbances in neuronal membranes, which leads to cell death (Twarowski and Herbet, 2023). A β aggregates can directly interact with the lipid and cholesterol components of the cell membrane, forming channels and destroying membrane integrity and permeability, which allows Ca²⁺ entry into the cell, leading to long-term potentiation (LTP) inhibition and neuronal death. Studies on neuroblastoma cell lines indicate impairments in tubulin and F-actin cytoskeleton, suggesting that this organelle is also very sensitive to A β oligomers (Jarero-Basulto et al., 2024).

Other reports suggest that neurodegeneration in AD cannot be solely attributed to insoluble A β fibrils. Growing evidence supports the amyloid- β oligomer (A β O) hypothesis, which posits that soluble A β Os are the primary toxic species driving pathogenesis (Wells et al., 2021). A β Os rapidly reduce synaptic receptor availability at the cell surface, leading to altered dendritic spine shape, decreased spine numbers, and synaptic damage in hippocampal neuron cultures. Studies using brain-derived A β Os reveal a complex mixture of soluble species contributing to synaptic toxicity. Evidence from Alzheimer's patient brain extracts indicates that low molecular weight, highly diffusible A β Os are particularly effective in disrupting both neuronal structure and synaptic function. These findings indicate that a small, bioactive subset of A β Os is primarily responsible for driving toxicity. Unlike amyloid plaques, A β Os

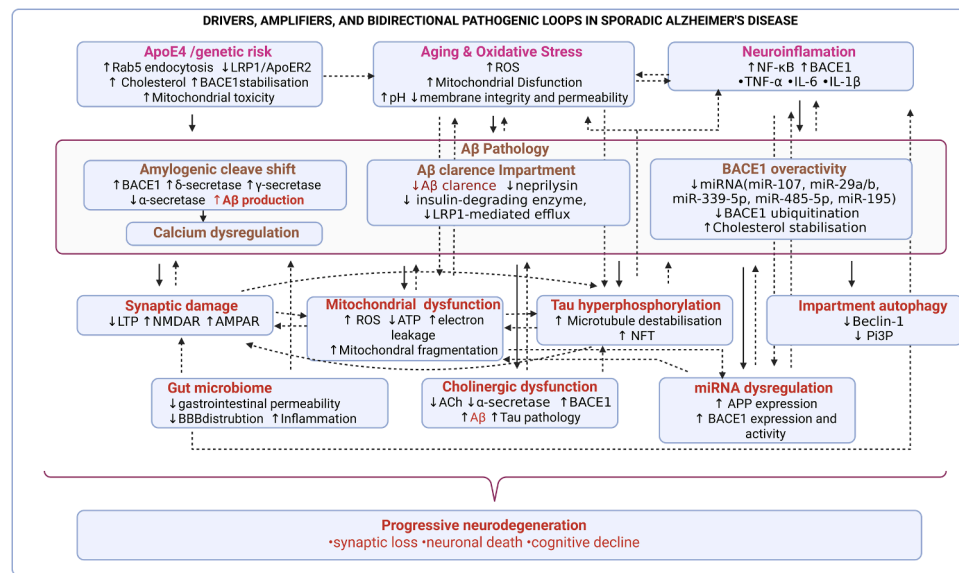


Fig. 3. Schematic overview of the interconnected pathways contributing to sporadic Alzheimer's disease. Genetic risk factors, particularly ApoE4, along with aging and oxidative stress, and neuroinflammation, initiate and exacerbate disease progression through effects on endocytosis, cholesterol metabolism, mitochondrial function, and inflammatory signaling. These upstream processes converge on amyloid- β ($A\beta$) pathology, including increased amyloidogenic APP processing, impaired $A\beta$ clearance (via neprilysin, insulin-degrading enzyme, and LRP1-mediated efflux), and enhanced BACE1 expression, activity and stabilization.

are especially detrimental to synaptic function, disrupting LTP—a process crucial for learning and memory (Hampel et al., 2021; Chen et al., 2017; Cline et al., 2018; Krafft et al., 2022; Chen et al., 2024).

Like other brain metabolites, $A\beta$ is cleared through multiple pathways, including enzymatic degradation by proteases, transport across the blood–brain barrier (BBB), bulk flow of interstitial fluid, and absorption into the cerebrospinal fluid (CSF). When $A\beta$ production is elevated and homeostasis is disrupted, especially under conditions where secretase activities are altered, $A\beta$ is generated in excess, overwhelming clearance mechanisms even if they are functioning properly. This imbalance promotes toxic $A\beta$ aggregation, contributing to the progression of neurodegenerative processes such as those seen in sAD. Further, when clearance mechanisms fail to function properly, a cumulative and exacerbated pathological effect escalates (Tarasoff-Conway et al., 2015; Yoon and Jo, 2012; Cai et al., 2023; Cheng et al., 2020). A delicate balance between the continuous production and effective clearance of $A\beta$ is essential to maintain homeostasis and prevent the formation of toxic, misfolded aggregates.

Further, the development of neurodegeneration involves a dynamic interplay between $A\beta$ and various other contributing factors.

One of the most significant is the formation of NFTs (Metaxas and Kempf, 2016), which are composed of intracellular aggregates of hyperphosphorylated tau protein (Guo et al., 2017; Wang and Mandelkow, 2016). Tau is a microtubule-associated protein predominantly expressed in neurons. Its binding domain plays a vital role in facilitating microtubule polymerization and stabilization, which is essential for maintaining the structural integrity of the neuronal cytoskeleton (Sayas and Ávila, 2021; Khan et al., 2020). Disturbances in calcium homeostasis, caused by $A\beta$ plaques accumulation, lead to a significant increase in its intracellular concentration. This modification reduces tau's affinity for microtubules, promoting the formation of NFTs. Tangles accumulate in the cytosol, impairing essential cellular processes such as synaptic transmission, axonal transport, and signal transduction, ultimately leading to gradual neuronal degeneration (Twarowski and Herbet, 2023; Guo et al., 2017; Khan et al., 2020).

Neurons are particularly susceptible to oxidative stress due to several intrinsic factors that contribute to elevated levels of reactive oxygen species (ROS) generation. Their high metabolic demand is primarily met through oxidative phosphorylation, a process that requires intense

mitochondrial activity for energy (Murali Mahadevan et al., 2021). This elevated mitochondrial function increases the likelihood of electron leakage, leading to the formation of superoxide radicals, which are subsequently converted into H_2O_2 by superoxide dismutase. Additionally, neurons—especially those within dopaminergic, noradrenergic, and serotonergic systems—express high levels of monoamine oxidase, an enzyme responsible for the breakdown of neurotransmitters. This enzymatic activity further contributes to H_2O_2 accumulation as a metabolic byproduct. Compounding this vulnerability, neurons possess relatively low levels of antioxidant defenses, such as catalase and glutathione peroxidase, when compared to other brain cells like astrocytes. As a result, their capacity to neutralize excess H_2O_2 is limited, increasing the risk of oxidative damage (Barnham et al., 2004; Lin and Beal, 2006).

The aggregation of $A\beta$ intensifies oxidative stress by facilitating the production of reactive oxygen species through several mechanisms, including interactions with metal ions such as iron and copper, which catalyze ROS generation via the Fenton reaction. Furthermore, $A\beta$ accumulation disrupts mitochondrial function, leading to increased electron leakage and ROS production, while also impairing cellular membranes and triggering chronic inflammation via microglial activation; this, combined with the depletion of antioxidant defenses, creates a vicious cycle that exacerbates neurodegeneration in Alzheimer's. All of these factors contribute to cell membrane damage, cytoskeletal changes, and ultimately neuronal cell death (Tamagno et al., 2021; Kowalczyk et al., 2021; Cheignon et al., 2018; Yu and Luo, 2024).

Downstream consequences include calcium dysregulation, synaptic damage (altered LTP, NMDA and AMPA receptor signaling), mitochondrial dysfunction (increased ROS, reduced ATP, electron leakage, and fragmentation), tau hyperphosphorylation leading to neurofibrillary tangle (NFT) formation, and impaired autophagy (reduced Beclin-1 and PI3P). Additional contributing factors include gut microbiome alterations, cholinergic dysfunction, and miRNA dysregulation affecting APP and BACE1 expression.

These processes are highly interconnected, forming feedback loops that amplify $A\beta$ and tau pathology, ultimately leading to progressive neurodegeneration characterized by synaptic loss, neuronal death, and cognitive decline.

Another critical factor in AD is cholinergic dysfunction. The

cholinergic system plays a vital role in learning and memory processes through acetylcholine (ACh) release at both muscarinic and nicotinic receptors (Berry and Harrison, 2023). In AD, there is a marked loss of cholinergic neurons in the basal forebrain, resulting in reduced levels of acetylcholine, a neurotransmitter essential for cognitive function. This decline in cholinergic activity correlates with the severity of cognitive impairment in individuals with AD (Hempel et al., 2018; Chen and Mobley, 2019). Studies in animals have indicated that cholinergic depletion accelerates A β accumulation and tau pathology, both of which are linked to cognitive decline (Ramos-Rodriguez et al., 2013). Under physiological conditions, ACh acting through M1 and M3 muscarinic receptors activates PKC, which simultaneously stimulates α -secretase (ADAM10/ADAM17) to favour non-amyloidogenic APP cleavage, phosphorylates BACE1 at serine 498 to reduce its catalytic activity and promote its lysosomal degradation, and inhibits presenilin-1 to reduce γ -secretase efficiency. Loss of cholinergic tone in sAD removes all three of these anti-amyloidogenic brakes simultaneously — reducing non-amyloidogenic processing while permissively elevating amyloidogenic processing in a coordinated fashion that operates additively with NF- κ B-driven, oxidative stress-driven, and ApoE4-driven BACE1 upregulation (Hempel et al., 2018; Campos-Peña et al., 2022).

In sporadic AD, APP dysregulation is thought to result primarily from altered activity of amyloidogenic secretases, trafficking and clearance rather than direct structural changes in APP. Increased BACE1 expression and activity have been reported in sAD brains, in contrast to fAD, where such changes are less prominent (Pera et al., 2013). This upregulation enhances β -secretase cleavage of APP, although accumulation of β -CTF is typically less pronounced in sAD compared to familial, which reflects contrast in processing dynamics. In addition to altered production, intracellular accumulation of A β 42 is a hallmark of sporadic forms of the disease. This accumulation may result either from increased amyloidogenic APP processing, leading to elevated β -CTF and A β 42 levels, or from impaired extracellular clearance followed by neuronal uptake of A β (Small et al., 2017). Indeed, impaired A β clearance is considered a central pathogenic mechanism in LOAD, involving reduced activity of degrading enzymes such as neprilysin and insulin-degrading enzyme, disrupted blood–brain barrier transport (e.g., decreased LRP1 and increased RAGE-mediated influx), and dysfunction of glymphatic pathways (Tarasoff-Conway et al., 2015). ApoE4 further exacerbates these processes by influencing lipid metabolism, endosomal trafficking, and synaptic function. It enhances Rab5-mediated endocytosis, promoting the accumulation of APP and BACE1 within early endosomes, where amyloidogenic processing is favored. Additionally, cholesterol, which interacts with APP β -CTF, modulates secretase accessibility; elevated cholesterol levels, together with ApoE4 expression, synergistically increase β -CTF and BACE1 levels, thereby amplifying A β production (Lai et al., 2021; Zhu et al., 2015).

Importantly, the A β generated through this dysregulated amyloidogenic processing, particularly in its soluble oligomeric form, contributes directly to mitochondrial dysfunction, oxidative stress, inflammatory activation of microglia and astrocytes, and synaptic injury (Bigi et al., 2022; Hempel et al., 2021). These downstream effects are closely interconnected and form self-amplifying pathogenic loops that further enhance neuronal damage. The relationship between A β , mitochondrial dysfunction, and oxidative stress represents one of the most mechanistically important and self-sustaining pathological axes in sporadic AD. Rather than a simple linear sequence, the three elements form a closed feed-forward loop in which each component amplifies the other two, progressively entrapping neurons in an escalating cycle of bioenergetic failure and amyloidogenic processing. A β 42 directly inhibits ETC Complex I and Complex IV, forcing electron leakage as superoxide, collapsing the mitochondrial membrane potential, and reducing ATP synthesis (Reddy and Beal, 2008; Wang et al., 2008). A β 42 simultaneously hyperactivates Drp1-mediated mitochondrial fission while impairing mitophagy, causing dysfunctional ROS-generating mitochondria to accumulate progressively (Fang et al., 2019).

NMDAR-driven calcium overload further overwhelms mitochondrial buffering, activating the mPTP through cyclophilin D sensitisation and initiating apoptotic signalling in severely affected neurons (Reddy and Beal, 2008). The feed-forward loop is closed when accumulated ROS activates NF- κ B, which directly upregulates BACE1 transcription — producing more A β that re-enters mitochondria to perpetuate the cycle — while extramitochondrial A β –metal ion Fenton reactions oxidatively inactivate BACE1-suppressing miRNAs and the GGA3 BACE1 degradation adaptor, amplifying BACE1 at post-transcriptional and post-translational levels simultaneously (Tamagno et al., 2021; Cheignon et al., 2018). Tau hyperphosphorylation deepens the loop by independently impairing Complex I and mitochondrial transport, establishing the A β –tau–mitochondria triad as the central self-amplifying engine of sAD neurodegeneration (Reddy and Beal, 2008).

Even though inflammation in the body is meant to be protective, an overly strong inflammatory response can lead to tissue damage and contribute to disease development. A β drives neuroinflammation through activation of microglia and astrocytes, resulting in the release of pro-inflammatory mediators that exacerbate neuronal injury and can further increase amyloidogenic processing. Neuroinflammation is not merely a downstream consequence of A β deposition, but part of a self-amplifying pathogenic circuit. Multiple studies suggest that neuroinflammation may act as an independent factor in AD, with cytokines like TNF- α , IL-6, IL-1 β playing a central role as well transcription factor NF- κ B (Thakur et al., 2023; Calsolaro and Edison, 2016). It takes part in processes such as amyloid deposition, neuronal damage, NFTs formation, and cell death (Calsolaro and Edison, 2016).

In addition, emerging evidence suggests that microRNAs participate in this network by regulating APP and BACE1 expression as well as inflammatory and synaptic signaling pathways. These mechanisms have important therapeutic implications. Unlike fAD, where reducing A β production may represent a more direct strategy, sAD likely requires broader approaches that also address impaired A β clearance, endosomal–lysosomal dysfunction, mitochondrial stress, neuroinflammation, synaptic damage, and regulatory pathways involving microRNAs. Accordingly, current strategies increasingly focus not only on immunotherapy-mediated A β removal, but also on modulation of ApoE function, lipid metabolism, and cellular homeostasis (Villain et al., 2022; Yang et al., 2023; Almeida et al., 2020). Regarding BACE1 pathology, at the post-transcriptional level, the loss of BACE1-suppressing microRNAs removes the normal brake on BACE1 mRNA translation. Under physiological conditions, miR-107, miR-29a/b, miR-339-5p, miR-485-5p, and miR-195 bind the BACE1 mRNA 3'-UTR and suppress its translation. In sAD, these miRNAs are downregulated through two parallel mechanisms: oxidative modification of guanosine residues within their seed regions impairs base-pairing fidelity, while DNMT3a-mediated epigenetic methylation of their gene promoters reduces transcriptional output. Additionally, ROS stabilises BACE1 mRNA by promoting HuD-mediated mRNA stabilisation, increasing its half-life and amplifying translational output beyond what transcriptional induction alone produces. The net effect is that BACE1 mRNA is simultaneously more abundant, more stable, and more efficiently translated in the sAD brain than in age-matched controls (Hébert et al., 2008; Rahman et al., 2024).

Furthermore, BACE1 protein stability is increased through impairment of its normal degradation machinery. Under physiological conditions, GGA3 ubiquitinates BACE1 and targets it for proteasomal degradation. In sAD, ROS oxidatively modifies GGA3, accelerating its own degradation and reducing BACE1 ubiquitination, thereby increasing steady-state BACE1 protein levels independently of any change in transcription or translation. Broader UPS impairment — caused by the accumulation of oxidatively damaged proteins that overwhelm proteasomal capacity — compounds this effect. Cholesterol accumulation, driven by ApoE4-mediated lipid dyshomeostasis, provides a further post-translational stabilisation mechanism by shifting

BACE1 into lipid raft microdomains where it is protected from lysosomal degradation (Tamagno et al., 2021).

At the synaptic level, soluble A β oligomers are strongly associated with synaptic impairment, including reduced dendritic spine density, impaired long-term potentiation (LTP), enhanced long-term depression (LTD), and disturbed glutamatergic neurotransmission, all of which correlate closely with cognitive decline. Mechanistically, A β oligomers inhibit hippocampal long-term potentiation (LTP), facilitate LTDN, and reduce synaptic efficacy through NMDA receptor-dependent pathways (NMDAR is hyperactivated in AD), and this is one of the most important distinctions from their normal physiological role.) and altered AMPAR trafficking (Walsh et al., 2002; Shankar et al., 2007; Shankar et al., 2008; Li and Selkoe, 2020).

What is more, emerging evidence suggests a link between gut microbiome composition and AD. The gut microbiota undergoes changes with aging, which not only results in gastrointestinal issues but also contributes to central nervous system disorders, including dementia (Kesika et al., 2021; Zhang et al., 2023). Numerous experimental and clinical studies have highlighted the impact of gut microbiota on cognition (Cooke et al., 2022; Ni Lochlainn et al., 2024), with its dysbiosis linked to neurodegeneration in aging. This imbalance triggers the release of amyloid and lipopolysaccharides, disrupting gastrointestinal permeability and the BBB. As a result, inflammatory signaling pathways are activated, promoting neuroinflammation, neuronal damage, and, ultimately, neuronal death in AD (Kesika et al., 2021).

Finally, autophagy, a cellular process responsible for degrading damaged organelles and proteins (Wang et al., 2019), is often impaired in AD. Especially in the early phase of disease, the buildup of A β may trigger autophagy, with the autophagosome-lysosome system playing a role in its breakdown. Research indicates that A β accumulates in defective autophagic vesicles, which may contribute to the development of extracellular A β plaques (Zhang et al., 2022). It has been shown that the core subunits of the vacuolar protein sorting 34 (VPS34) complex (important node of autophagosome generation), particularly Beclin-1 (key protein in autophagy), are significantly reduced as AD progresses (Levine and Kroemer, 2019). Consistent with this, the production of PI3P, which is regulated by the VPS34 complex, is also found to be decreased in the brains of AD patients. As a result, it is thought that the buildup of autophagic structures, such as autophagosomes and autolysosomes, occurs due to a blockage in the autophagic flux (Zhang et al., 2021; Reddy and Oliver, 2019).

A key challenge for therapeutic development is to understand why APP processing shifts toward the amyloidogenic pathway in sporadic AD and what drives secretase activity under pathological conditions. Increasing evidence indicates that this shift is not caused by structural abnormalities in APP or secretases, but rather by changes in their localization, trafficking, and regulatory mechanism. Factors such as altered membrane lipid composition, particularly increased cholesterol and lipid raft formation, promote the colocalization of APP with secretases, thereby favoring amyloidogenic processing. In addition, endosomal-lysosomal dysfunction enhances APP internalization and exposes it to acidic compartments where some of the secretases activity is optimal. Further, restoring mitochondrial function represents a particularly critical therapeutic priority in sporadic AD, given that mitochondrial dysfunction occupies a uniquely central position in the disease network — simultaneously driving A β overproduction through ROS-mediated NF- κ B/BACE1 upregulation, accelerating tau hyperphosphorylation through energy deprivation and calcium dysregulation, amplifying neuroinflammation and directly causing the synaptic bioenergetic failure that underlies cognitive decline. The systematic nature of miRNA loss in sAD — multiple miRNAs targeting the same mRNA, all downregulated through the same upstream pathological drivers — suggests that miRNA replacement strategies face a fundamental challenge: restoring any single miRNA is unlikely to fully restore BACE1 suppression if the remaining network members remain downregulated. This argues for upstream intervention — reducing the oxidative stress

and neuroinflammation that drive miRNA loss across the entire network — in combination with targeted miRNA restoration for the most clinically impactful members. miR-107 and miR-298, which simultaneously target both BACE1 and APP mRNAs, represent particularly high-priority candidates because their restoration addresses substrate and enzyme elevation simultaneously.

Sporadic Alzheimer's disease is not a uniform disorder but a highly individualised pathological landscape in which the relative contribution of each disease axis — amyloidogenic APP processing, mitochondrial dysfunction, neuroinflammation, oxidative stress, synaptic damage, cholinergic loss, and miRNA dysregulation — varies substantially between patients according to their unique genetic background, metabolic history, and aging trajectory. Consequently, a precision medicine approach guided by comprehensive biomarker profiling might be essential for enabling the rational selection and proportioning of combination therapeutic strategies matched to the dominant pathogenic drivers in each individual rather than applying a uniform intervention to a fundamentally heterogeneous disease (Jack et al., 2024; Cummings et al., 2024; Villain et al., 2022).

3. APP-processing pathways

As shown above A β accumulation disrupts numerous cellular processes, including energy metabolism, calcium homeostasis, membrane integrity, and synaptic signaling, ultimately contributing to widespread neuronal dysfunction. Therefore, gaining a precise understanding of the biochemical and enzymatic mechanisms underlying A β generation is crucial for the development of effective therapeutic strategies. While the idea of suppressing A β production appears to be a logical and promising approach, it cannot be successfully achieved without a deep knowledge of the complex chemistry and molecular pathways involved in its formation which arises from the proteolytic processing of APP by a series of proteolytic enzymes.

3.1. Canonical APP processing (non-amyloidogenic pathway and amyloidogenic pathway, Fig. 4)

APP processing involves three key proteases: α -, β -, and γ -secretases (Chen, 2015) that regulate two distinct pathways (Haass et al., 2012; Zhao et al., 2020). In the amyloidogenic pathway (Fig. 4), β - (EC: 3.4.23.46) and γ -secretases (EC: 3.4.22.34) act sequentially, leading to the production of A β (van Maanen et al., 2018). In contrast, the non-amyloidogenic pathway (Fig. 4) involves cleavage of APP by α -secretase (EC: 3.4.24.81) (Azargoonjahromi, 2024), either producing soluble APP α (sAPP α) directly or generating shorter A β fragments, such as A β 1–15 and A β 1–16, through the combined action of β - and α -secretases. These two pathways yield different by-products with unique functional properties, potential physiological roles, and varying implications for disease processes (Hampel et al., 2021).

In the amyloidogenic pathway, the first step is carried out by β -secretase (Fig. 4), also known as β -site APP-cleaving enzyme-1 (BACE1) (Bhatia et al., 2023). It belongs to the aspartic protease family, and cleavage of APP770 between M671 and D672 amino acids (Fig. 4, Table 1) (Masi et al., 2023; Dislich and Lichtenthaler, 2012) results in the extracellular release of the soluble APP β (sAPP β) fragment and leaves behind the membrane-bound C99 domain (Hampel et al., 2021).

Beyond the M671-D672 cleavage site, BACE1 is also able to cleave APP770 within the A β domain at Y681–E682, which is also called the β' site (Zhang et al., 2017). This produces nonamyloidogenic C89 and subsequently generates amino-terminally truncated A β species, namely A β 11–40/42. Under physiological conditions, the Y681–E682 site serves as the primary β -cleavage site, and A β 11–40/42 is the predominant form detected in neuronal cultures (not presented in the figure). Both A β 11–40/42 and p3 are also present in cerebrospinal fluid and in conditioned media from cultured cells.

The choice between M671-D672 and Y681–E682 cleavage by BACE1

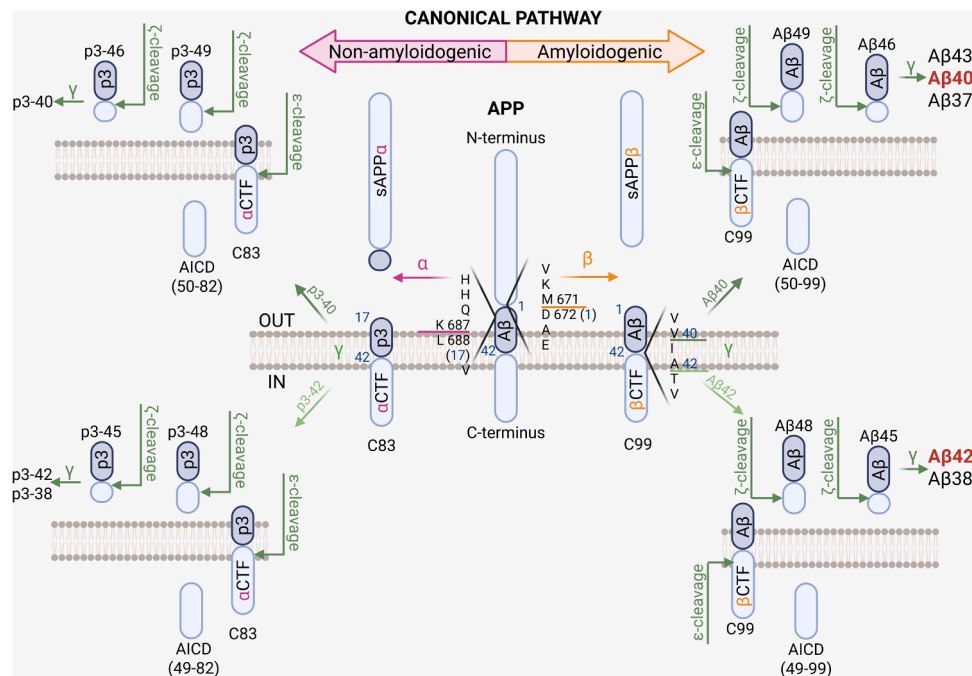


Fig. 4. Schematic representation of canonical amyloidogenic and non-amyloidogenic pathways of APP770 processing. In the non-amyloidogenic pathway, APP is first cleaved by α -secretase within the A β region, preventing A β formation and generating sAPP α and a C83, which is further processed by γ -secretase. In contrast, the amyloidogenic pathway involves initial cleavage by β -secretase, producing sAPP β and a C99 fragment. Subsequent cleavage of C99 by γ -secretase leads to the release of A β peptides, which can aggregate into extracellular plaques. Created with BioRender.com.

is strongly influenced by the local APP sequence near the β -cleavage sites and by the subcellular distribution of BACE1, rather than by factors such as organelle pH, glycosylation, or substrate levels. When BACE1 was engineered with a KKXXX ER retrieval motif on its cytoplasmic tail, it was retained in the ER in an immature state, which favored cleavage at M671-D672. By contrast, replacing the BACE1 cytoplasmic tail with the intracellular domain of murine furin redirected the enzyme to the trans-Golgi network, where it preferentially generated C89 instead of C99. In its wild-type form, BACE1 localizes mainly to the trans-Golgi network and endosomal system, where C89 production exceeds that of C99. Targeting BACE1 to lipid rafts through a glycosylphosphatidylinositol anchor enhanced its expression at the cell surface and promoted M671-D672 cleavage, increasing secretion of full-length A β . Notably, this shift in cleavage preference was independent of APP localization or the KM/NL mutation. BACE1 can also process C99 further at Y681-E682 to produce C89; however, APP-C99 engineered with the KK motif in the ER yielded little C89 compared to wild-type APP-C99. Even more unexpectedly, this construct produced very little A β overall, despite colocalizing with PS1 in the ER. A β generation from C99-KK could be partially restored by treatment with brefeldin A.

Together, these findings indicate that C99 and C89 are processed in distinct subcellular compartments: C99 is primarily produced in the ER by immature BACE1, whereas C89 is mainly generated in downstream organelles, including the trans-Golgi network and lysosomal system (Zhang and Song, 2013; Deng et al., 2013; Kimura et al., 2016; Huse et al., 2002).

BACE1 employs a general acid-base catalytic mechanism to cleave peptide bonds during its enzymatic cycle (Barman and Prabhakar, 2013). In this process, the D228 residue acts by abstracting a proton from a water molecule, thereby generating a hydroxide ion (OH⁻). This hydroxyl ion then performs a nucleophilic attack on the carbonyl carbon of the target peptide bond, and concomitantly D32 donates its proton to the carbonyl oxygen of the peptide bond to produce the tetrahedral gem-diol intermediate. The two aspartates reverse their functions and D32 now acts as a base while D228 acts as an acid. The protonated D228 donates its proton to the amide nitrogen of the scissile peptide bond, and

in a synchronous manner D32 abstracts a proton from one of the hydroxyl groups of the gem-diol intermediate (Table 1). This process leads to the cleavage of the peptide bond that generates the amino and carboxyl terminals (Barman and Prabhakar, 2013).

BACE1 is a membrane-associated enzyme with complex cellular trafficking (Kang et al., 2012). It is synthesized in the ER (Zhang and Song, 2013) as a proenzyme (zymogen) and undergoes post-translational modifications in the Golgi to gain full activity. The zymogen form exhibits considerable proteolytic activity, which approximately doubles upon prodomain removal. Instead of inhibiting enzymatic activity, the prodomain aids in the proper folding of the protease domain (Dislich and Lichtenthaler, 2012). Once BACE1 is inserted into the plasma membrane's lipid rafts, it can be released extracellularly (Motoki et al., 2012), but APP processing by BACE1 rarely occurs at the plasma membrane. Instead, extracellular BACE1 is typically endocytosed and transported to endosomes, where it becomes functionally active or sent to lysosomes for degradation (Fourriere et al., 2022).

Within the cell, BACE1 is predominantly found on the membranes of the trans-Golgi network and endosomes (Huse et al., 2002), where APP processing primarily takes place (Orobets and Karamyshev, 2023). Interestingly, while APP is internalized from the plasma membrane via a clathrin-mediated mechanism, BACE1 follows a clathrin-independent route (Aow et al., 2023). BACE1's optimal activity occurs at a pH of 5.5 (Prasad and Rao, 2023), making endosomes and lysosomes the primary sites of APP processing and A β production. APP and BACE1 eventually colocalize in Rab5-positive endosomes, where APP is cleaved by fully active BACE1 (Orobets and Karamyshev, 2023).

The second pathway is initialized by α -secretase cleavage at the α -site, located between residues K687 and L688 (Fig. 4, Table 1), within the plasma membrane (Yan et al., 2017). It occurs within the A β region of APP770, thereby preventing the formation of A β (Kummer and Heneka, 2014). As a result, the process generates sAPP α , a secreted peptide released into the extracellular space, and a membrane-bound C-terminal fragment known as C83 (or α -CTF) (Masi et al., 2023). In disintegrin and metalloproteinase domain-containing protein 10's

Table 1

Summary of proteases involved in the processing of APP770. The table lists proteases grouped by enzyme class, including their alternative names, APP cleavage sites, catalytic mechanisms, Ensembl identifiers, EC numbers, and chromosomal locations, providing an integrated biochemical and genomic overview relevant to AD.

Name (group)	Other names and symbols	Cleave site	Catalytic mechanism	Ensembl and EC number	Chromosomal location
α -secretase (metallo-endoprotease)	a disintegrin and metallo-proteinase domain-containing protein 10, ADAM10, CDw156, CD156c	K687- L688	Water activated by zinc ion acts as a nucleophile, E384 abstracts a proton from the zinc-bound water molecule, generating a OH ⁻ , which is a strong nucleophile. attacking the carbonyl carbon of the peptide bond to be cleaved (Seegar et al. 2017).	ENSG00000137845, EC: 3.4.24.81	Chromosome 15: 58,588,809–58,749,791 reverse strand. GRCh38:CM000677.2
β -secretase (aspartic-endoprotease)	β -site APP-cleaving enzyme-1, BACE1, membrane-associated aspartic protease 2, memapsin-2, aspartyl protease 2, ASP2	M671-D672 Y681–E682	Water activated by D228 acts as a nucleophile and attacks the peptide bond, aided by D32 proton donation. After forming a tetrahedral intermediate, their roles reverse, D32 abstracts a proton, and D228 donates one, leading to peptide bond cleavage and formation of amino and carboxyl termini (Barman and Prabhakar, 2013).	ENSG00000186318, EC:3.4.23.46	Chromosome 11: 117,285,232–117,316,259 reverse strand. GRCh38: CM000673.2
γ -secretase (aspartic endoprotease)	gamma-secretase activating protein consists of four individual proteins: PSEN1 (presenilin-1) nicastrin, APH-1 (anterior pharynx-defective 1), and PEN-2 (presenilin enhancer 2)	A β 40 product line: - ϵ -cleavage: T719-L720 - ζ -cleavage: I716-V717 - γ -cleavage: G709-V710, A713-T714 A β 42 product line: - ϵ -cleavage: L720-V721 - ζ -cleavage: V717-I718 - γ -cleavage: T714-V715, V711-I712, G708-G709	Nucleophilic attack via activated water similar to β -secretase mechanism, D257 activates the water for nucleophilic attack, D385 stabilizes transition state / assists in proton transfer (Masi et al., 2023; Wolfe and Miao, 2022; Wu et al., 2024).	ENSG00000186088, EC:3.4.22.34	PSEN1: Chromosome 14: 73,136,418–73,223,691 forward strand. GRCh38:CM000676.2 Nicastrin: Chromosome 1: 160,343,294–160,358,952 forward strand. GRCh38: CM000663.2 APH-1: Chromosome 1: 150,265,399–150,269,580 reverse strand. GRCh38: CM000663.2 PEN-2: Chromosome 19: 35,745,678–35,754,519 forward strand. GRCh38:CM000681.2
δ -secretase (cysteine protease)	legumain, asparagine endopeptidase, AEP	For APP695: N373-E374, N585-I586 For APP770: N448-E449 N660-I661	H148 deprotonates C189, generating a reactive thiolate anion (RS ⁻), which attacks the carbonyl carbon of the scissile bond, forming a tetrahedral intermediate. Its collapse releases the C-terminal fragment. Water, also activated by H148, cleaves the acyl-enzyme intermediate, releasing the N-terminal fragment and restoring the enzyme (He et al., 2020; Vandooren et al., 2016; Dall and Brandstetter, 2013). Similar to ADAM10, it uses a zinc-bound water molecule, activated by a nearby glutamate, to generate a OH ⁻ .	ENSG00000100600, EC: 3.4. 22.34	Chromosome 14: 92,703,807–92,748,679 reverse strand. GRCh38:CM000676.2
meprin β (metallo-endoprotease)	astacin metalloprotease meprin β , MEP1B, endopeptidase-2, N-benzoyl-L-tyrosyl-P-amino-benzoic acid hydrolase subunit beta, PABA peptide hydrolase, PPH beta	M671-D672, D672-A673, A673-E674, S124-D125, E455-T456, G458-D459	This ion performs a nucleophilic attack on the carbonyl carbon of the peptide bond, forming a zinc-stabilized tetrahedral intermediate that collapses, leading to bond cleavage and fragment release (Cui et al., 2017). Similar to ADAM10 and meprin β it uses a zinc-bound water molecule, it is a zinc-dependent protease that cleaves peptide bonds by activating a water molecule through a conserved glutamate residue, likely D204. The resulting OH ⁻ becomes the nucleophile that attacks the carbonyl carbon of the scissile	ENSG00000141434, EC: 3.4.24.63	Chromosome 18: 32,185,069–32,220,404 forward strand. GRCh38:CM000680.2
η -secretase (metallo-proteinase)	membrane-type 5-matrix metalloproteinase, MT5-MMP, MMP-24	N579-M580		ENSG00000125966	Chromosome 20: 35,226,690–35,276,998 forward strand. GRCh38:CM000682.2

(continued on next page)

Table 1 (continued)

Name (group)	Other names and symbols	Cleave site	Catalytic mechanism	Ensembl and EC number	Chromosomal location
θ secretase (aspartic endoprotease)	ASP1, BAE2, DRAP, AEPLC, ALP56, ASP21, CDA13, CEAP1	F690-F691 F691-A692	peptide bond in APP (Baranger et al., 2016; Yang et al., 2024). Similar to δ-secretase the active site centers on H121 and C163, where H121 stabilizes the substrate's carbonyl group and Cys163 serves as the nucleophile cleaving the peptide bond. C163 and G238 also help stabilize the tetrahedral transition state via hydrogen bonding interactions (Wyllie, 1997). The catalytic serine (S185) functions as a nucleophile. Upon activation by the histidine residue (H174), the hydroxyl group of S185 attacks the carbonyl carbon of the scissile peptide bond in the substrate. This results in the formation of a tetrahedral intermediate (Knopf et al. 2020).	ENSG00000182240.16 EC:3.4.23.45	Chromosome 21: 41,167,801–41,282,530 forward strand. GRCh38:CM000683.2
caspases-3 (cysteine-aspartic acid protease)	apoptosis-related cysteine protease 3, CASP3, Apopain Cysteine protease CPP32 (CPP-32) Protein Yama SREBP cleavage activity 1 (SCA-1)	D739-A740		ENSG00000164305, EC: 3.4.22.56	Chromosome 4: 184,627,696–184,650,062 reverse strand.GRCh38: CM000666.2
RHBDL4 (serine protease)	Rhomboid domain-containing protein 1 Rhomboid-like protein 4	M580-I581, G589-N590		ENSG00000144468, EC: 3.4.21.105	Chromosome 2: 226,835,581–226,999,215 forward strand. GRCh38:CM000664.2

(ADAM10) catalytic mechanism, a water molecule—activated by the zinc ion and a catalytic glutamate residue—is deprotonated to form a OH⁻, which then performs a nucleophilic attack on the carbonyl carbon of the peptide bond. This activated water molecule acts as the nucleophile and attacks the carbonyl carbon of the scissile peptide bond, leading to bond cleavage and substrate hydrolysis. High-resolution structural analyses have provided detailed views of ADAM10's active site. The catalytic domain features a five-stranded β-sheet and a central helix containing the HExxH motif. This motif includes two histidines (H383 and H387) coordinating the zinc ion, and a glutamate (E384) involved in catalysis. A third histidine (H393) completes the zinc coordination (Seegar et al., 2017). ADAM10 is the primary physiological α-secretase in the brain (Müller et al., 2017). This enzyme is synthesized in the ER as an inactive zymogen, comprising C-terminal, transmembrane, cysteine-rich, disintegrin, and zinc-binding metalloprotease domains (Brummer et al., 2018; Seegar et al., 2017). It also contains a pro-domain that keeps the protease inactive through a cysteine switch mechanism (Pastore et al., 2023), where a cysteine residue in the pro-domain binds to the zinc ion at the catalytic site, inhibiting proteolytic activity. As ADAM10 is transported to the plasma membrane, it undergoes N-glycosylation at four positions and matures through the removal of its pro-domain by proprotein convertases (Manzine et al., 2019).

The next step in both pathways is performed by the same enzyme—γ-secretase (Fig. 4) that cleaves the CTFs (Matsushima et al., 2012; Wolfe, 2019; Wu et al., 2024). The process involves a series of cleavage events, starting with an endoproteolytic cut at the ε-site (Xu, 2009), which releases the AICD, followed by consecutive exoproteolytic cuts moving toward the N-terminus. This sequence of cleavages results in the production of AICD along with either Aβ or p3 (Müller et al., 2017). γ-Secretase is a transmembrane protein complex composed of presenilin (PS), nicastrin, anterior pharynx defective-1 (Aph-1), and presenilin enhancer-2 (Pen-2) (Bai et al., 2015). It belongs to a unique class of proteases known as intramembrane-cleaving proteases (Kamp et al., 2015) and carries out an unusual cleavage process, targeting substrates within the lipid bilayer of membranes (Hur, 2022). The most important part of γ-secretase is PS, which plays the role of a catalytic component. Two fully conserved transmembrane aspartic acid residues are essential

for γ-secretase's proteolytic activity (Wolfe, 2019; Carter et al., 2008), classifying it within the aspartyl protease family. However, despite this classification, it is structurally distinct from the more familiar soluble aspartyl proteases. Presence of those two residues is essential for Aβ production (Wolfe, 2019). The proteolytic cleavage performed by γ-secretase can be categorized into sequential γ-, ζ-, and ε-cleavage sites (Checler et al., 2021), which include five γ-sites, two ζ-sites, and two ε-sites (Masi et al., 2023). This distinction is crucial for generating various Aβ species, as the different cleavage events by γ-secretase lead to the production of distinct Aβ peptides (Steiner et al., 2018). β-CTFs undergo ε-cleavage, resulting in either Aβ49 and the AICD 50–99 or Aβ48 and AICD 49–99. For the Aβ40 product line, Aβ49 is cleaved at the ζ-site to produce Aβ46, and further processed at the γ-sites to yield Aβ43, Aβ40, and Aβ37 through a tripeptide trimming process. In contrast, in the Aβ42 product line, Aβ48 is cleaved at the ζ-site to generate Aβ45 and then at the γ-sites to produce Aβ42 and Aβ38 through a tetrapeptide trimming process (Masi et al., 2023; Bolduc et al., 2016; Szaruga et al., 2017). It is Aβ42 that is responsible for the development of AD (Sturchio et al., 2021; Habchi et al., 2017; Sun et al., 2015). The key factor is its amino acid sequence, which causes neuronal toxicity, the ability to accumulate, form plaques, and neurodegeneration (Marshall et al., 2016). Moreover, knockout of γ-secretase components leads to embryonic lethal phenotypes that closely resemble those observed with Notch receptor knockouts. This indicates that γ-secretase's role in Notch signaling may be its most crucial biological function and likely the primary driver behind the evolution of this protease complex (Wolfe, 2020).

Although γ-secretase and β-secretase share a common catalytic strategy (nucleophilic attack via activated water), γ-secretase operates in a much more hydrophobic, membrane-embedded environment, making it structurally and functionally more complex. γ-Secretase, an intramembrane aspartyl protease, catalyzes peptide bond hydrolysis within the lipid bilayer. The active site contains two conserved aspartate residues, typically D257 and D385 (Table 1) in presenilin-1, that coordinate a water molecule. One of these aspartates activates the water by abstracting a proton, generating a OH⁻. This hydroxyl ion then performs a nucleophilic attack on the carbonyl carbon of the target peptide. This activated water molecule acts as the nucleophile, attacking the

carbonyl carbon of the scissile peptide bond in the substrate. This nucleophilic attack leads to the formation of a tetrahedral intermediate, which then collapses, resulting in cleavage of the peptide bond and release of the product (Wolfe and Miao, 2022; Wu et al., 2024).

3.2. Non-canonical APP processing

3.2.1. δ -secretase pathway

Noncanonical processing of APP by δ -secretase (EC: 3.4.22.34), also known as asparagine endopeptidase (AEP) or legumain (Poreba, 2019), represents an alternative cleavage pathway in AD pathology, separate from the more widely known amyloidogenic and non-amyloidogenic pathways (Fig. 5). δ -Secretase has gained attention for its role in proteolytically processing both APP and tau, two key proteins implicated in AD (Qian et al., 2024). The enzyme cleaves APP695 at two specific asparagine sites N373-E374 and N585-I586 (Fig. 5, Table 1) (Meng et al., 2025; English et al., 2021), yielding several fragments: three soluble ectodomain fragments (sAPP1–373, sAPP1–585, and sAPP374–585) and a δ -CTF (for APP770 the cleavage sites are N448-E449 and N660-I661 respectively). δ -CTF is further processed by β - and γ -secretases (Fig. 5), resulting in the production of A β peptides, which contribute to amyloid plaque formation, a hallmark of AD. Notably, δ -secretase activity and expression increase with age, aligning with the heightened risk of AD in older adults (Masi et al., 2023; Wang et al., 2021).

In addition to A β production, δ -secretase contributes to AD pathology through several interconnected mechanisms. The enzyme cleaves tau at asparagine residues N255 and N368 (Schlegel et al., 2019), producing tau fragments that lack the ability to stabilize microtubules, leading to tau hyperphosphorylation and aggregation into NFTs. The tau 1–368 fragment generated by δ -secretase is neurotoxic, exacerbating neuronal dysfunction and death (Kang et al., 2020). Deletion of δ -secretase in AD mouse models (e.g., 5xFAD and P301S tau models) results in reduced A β accumulation, decreased plaque and tangle formation, and improved cognitive function, highlighting δ -secretase's pathological role in both A β and tau pathologies (Yao et al., 2021).

Furthermore, δ -secretase-mediated cleavage of the APP fragment C586–695 promotes neuroinflammation through interaction with the

CCAAT/enhancer-binding protein beta (C/EBP β) transcription factor (Luo et al., 2025). This interaction drives C/EBP β nuclear translocation and enhances its transcriptional activity, leading to upregulated expression of genes associated with AD, including APP, MAPT (encoding tau), and LGMN (encoding δ -secretase itself), as well as inflammatory cytokines. This feedback loop intensifies A β production, tau pathology, and neuroinflammation, thereby worsening AD symptoms and disease progression (Masi et al., 2023; Yao et al., 2021; Yao et al., 2024).

AEP is regulated by pH (Zhang et al., 2015), activated in acidic environments characteristic of cellular stress, and found predominantly in endolysosomes. It is synthesized as inactive proenzymes within the ER and transported through the Golgi apparatus to the endosome. Activation takes place upon reaching the acidic environment of the lysosome. It also contains a caspase-like catalytic domain and a C-terminal legumain stabilization and activity modulation domain, separated by an activation peptide. The catalytic function of AEP relies on key residues—histidine (H148), cysteine (C189), and asparagine (N42)—in the catalytic domain, which together enable their unique hydrolytic activity. AEP belongs to the CD enzyme family, characterized by the HDAC catalytic motif, which they share with other proteases like caspases, gingipains, and separases.

The catalytic cysteine residue (C189 in human AEP) is activated by an adjacent histidine residue (H148). The histidine acts as a general base, abstracting a proton from the thiol group of cysteine, generating a more reactive thiolate anion (RS^-). The activated thiolate anion performs a nucleophilic attack on the carbonyl carbon of the scissile peptide bond, leading to the formation of a tetrahedral intermediate. Further, it collapses, resulting in the cleavage of the peptide bond and release of the C-terminal fragment. Subsequently, a water molecule, activated by the histidine residue, hydrolyzes the acyl-enzyme intermediate, releasing the N-terminal fragment and regenerating the free enzyme (He et al., 2020; Vandooren et al., 2016; Dall and Brandstetter, 2013).

AEPs' function is similar to that of cysteine cathepsins; both reside primarily in endolysosomal compartments and require low pH for activation and function. Despite lacking sequence homology, AEPs and cathepsins share similar protein degradation capabilities, making it challenging to design selective inhibitors that distinguish AEPs from cathepsins or caspases (Song, 2022; Zhang et al., 2014; Dall and

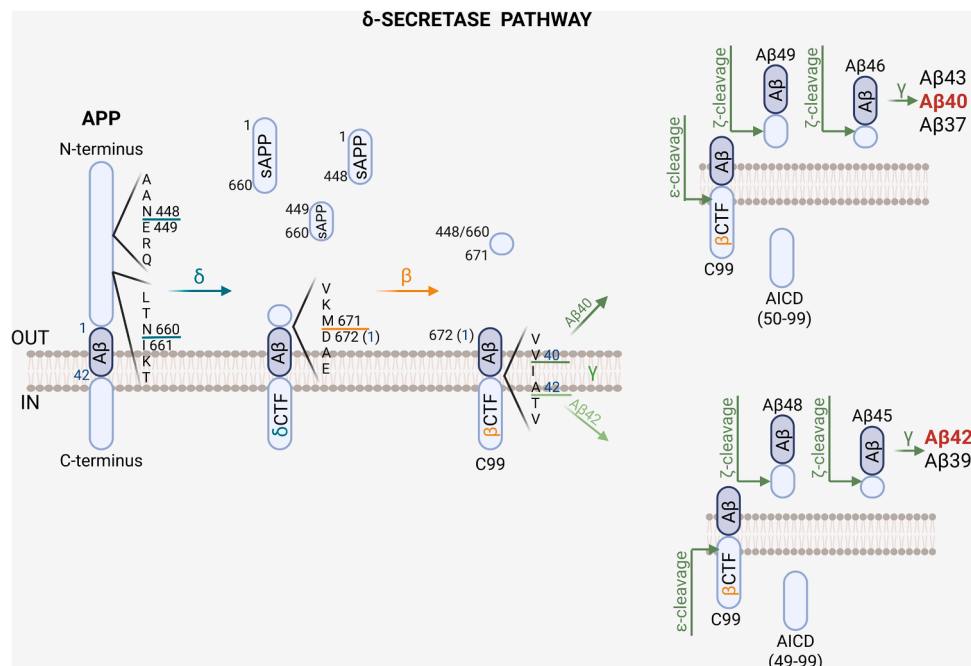


Fig. 5. Schematic representation of the amyloidogenic pathway of the APP770 processing via δ -secretase. δ -secretase cleaves APP at specific asparagine residues, facilitating further cleavage by β -secretase and enhancing A β generation. Created with BioRender.com.

Brandstetter, 2016; Elsässer et al., 2017).

Not only pH is vital in its regulation but also phosphorylation. For instance, serine-arginine protein kinase 2 phosphorylates δ -secretase at residue S226, facilitating its activation and cytoplasmic translocation, while brain-derived neurotrophic factor (BDNF)-activated Akt phosphorylates δ -secretase at T322, sequestering it in lysosomes and suppressing its activity. BDNF/TrkB pathway deficiencies, common in AD, can thus increase δ -secretase activity, further exacerbating APP and tau pathology (Yao et al., 2021; Xiang et al., 2019; Barbereau et al., 2020; Liao et al., 2021; Wang et al., 2019).

Interestingly, recent findings reveal that inflammatory cytokines activate the JAK2/STAT3 signaling pathway, which, in turn, upregulates C/EBP β expression and δ -secretase activity. This pathway establishes a molecular link between chronic neuroinflammation and δ -secretase activation, identifying δ -secretase as a central player in AD's complex pathology. Inhibiting δ -secretase has shown therapeutic potential in AD models, with inhibitors like CP11 reducing A β production, tau aggregation, and neuroinflammation, thereby preserving synaptic function and cognitive abilities (Wang et al., 2019; Liao et al., 2021; Yang et al., 2017). Given its multifaceted involvement in AD, δ -secretase remains a promising drug target, and ongoing research aims to further elucidate its role and refine therapeutic strategies for AD intervention (Zhang et al., 2024).

3.2.2. Meprin β pathway

BACE1 specifically generates A β peptides that start at positions 1 or 11 (A β 1-x/11-x), meaning that A β species beginning at other N-terminal sites, such as position 4 (A β 4-x) or 2 (A β 2-x), cannot be attributed to BACE1 activity. The presence of these alternative A β forms without BACE1 suggests an independent APP processing pathway involving other β -secretases (Marengo et al., 2022; Bien et al., 2012; Kimura et al., 2016; Armbrust et al., 2024). The zinc-dependent astacin metalloprotease meprin β (EC: 3.4.24.63) is a potential candidate for directly generating N-terminally truncated A β while BACE1 is inactive (Becker-Pauly and Pietrzik, 2017; Wichert et al., 2019). This particular family of proteases is characterized by multidomain extracellular proteins with a conserved core structure, including a signal peptide for secretion, a prodomain or prosegment, and a zinc-dependent catalytic domain (Gomis-Rüth and Stöcker, 2023). Its catalytic activity is centered around a zinc ion located in the enzyme's active site, which is coordinated by histidine residues within a conserved HEXxH motif, along with an adjacent glutamate residue (Cui et al., 2017). The catalytic mechanism of meprin β proceeds through a nucleophilic attack initiated by a water molecule (Table 1). This water molecule is bound to the zinc ion and becomes polarized through interactions with the adjacent glutamate residue, which acts as a general base. As a result, the water molecule is deprotonated and converted into a more reactive hydroxide ion. The activated hydroxide ion then performs a nucleophilic attack on the carbonyl carbon of the scissile peptide bond in the substrate. This leads to the formation of a tetrahedral intermediate, which is stabilized by the zinc ion. The intermediate subsequently collapses, resulting in the cleavage of the peptide bond and release of the hydrolyzed peptide fragments (Nagase, 2013; Dumermuth et al., 1991; Gomis-Rüth, 2009; Gomis-Rüth, 2009). This catalytic strategy is characteristic of astacin-like metalloproteases and supports meprin β 's role in diverse physiological processes, including extracellular matrix remodeling and inflammation (Tanase et al., 2025).

Meprin- β proteolytically cleaves at the APP695 β -site M596-D597 and also targets the adjacent sites D597-A598 and A598-E599. What is more, three additional cleavage sites within APP770 have been identified between S124-D125, E455-T456, and G458-D459 (Fig. 6). However, subsequent *in vitro* analysis revealed an N-terminal proteolytic fragment with a molecular mass of 11 kDa, which resulted from cleavage at S124-D125 and directly matched a fragment found in human CSF. The subcellular location of APP may determine the site of meprin β 's cleavage; membrane-bound meprin β cleaves APP at the β -site, while

soluble meprin β yields the N-terminal APP fragments. This process produces fragments similar to sAPP β (sAPP β *) and two shorter soluble fragments (sAPP1-380/3 and sAPP1-124) (Andrew et al., 2016; Jefferson et al., 2011). Additionally, a CTF known as C99* or CTF β *, which is one amino acid shorter than C99 (that is produced in the canonical pathway, where BACE1 is involved) remains in the cell membrane. When γ -secretase cleaves C99*, it leads to the formation of N-terminally truncated A β peptides (A β 2-40/42) that are highly prone to aggregation (Menon et al., 2019). Interestingly, following β -secretase inhibition, a strong signal for A β 1-40 remains detectable, along with a prominent amount of the amino-terminal truncated A β 2-40 variant and additional truncated A β variants. These findings suggest that, even in the absence of β -secretase activity, meprin β can continue to produce A β 1-40, A β 1-39, and A β 2-40. Overall, this indicates that meprin β has the capacity to generate A β 1-40, effectively mimicking β -secretase-like activity (Bien et al., 2012; Schönherr et al., 2016).

The significant difference between these two enzymes is the localization in the cell where cleavage takes place. In contrast to BACE1, meprin β performs cleavage of APP at the cell surface and may directly compete with ADAM10 *in vivo*. This is evidenced by the elevated levels of sAPP α found in the soluble fractions from the brains of meprin β knockout mice. Furthermore, the absence of meprin β also led to an increase in sAPP β within the soluble brain fraction, as shown by a neoeptide-specific antibody. Additionally, conditioned medium from cells expressing the APPSWE variant—which is a more efficient substrate for BACE1 than the wild-type APP—significantly changed the production ratio of A β 2-40 to A β 1-40, suggesting that meprin β and BACE1 compete directly for APP (Schönherr et al., 2016). What is more, meprin β has the ability to activate ADAM10 by cleaving its pro-peptide, and in response, ADAM10 can release the inactive form of meprin β from the plasma membrane. This interaction suggests that the processing of APP by ADAM10 and meprin β is regulated by a feedback loop, which helps control the activity of these two metalloproteases (Scharfenberg et al., 2019; Li et al., 2022). There are studies suggesting that particular APP posttranslational modification, namely phosphorylation of serine in position 675, promotes meprin β -mediated and decreases α -secretase-mediated processing of APP at the plasma membrane. However, the role of this modification still has to be further investigated (Menon et al., 2019; Armbrust et al., 2021).

3.2.3. η -pathway

To identify novel proteolytic pathways for APP, a focus was placed on the main differences in molecular mass, particularly for CTFs that do not lead to the generation of the P3 fragment (CTF- α) or amyloid β (CTF- β). An investigation revealed a new CTF with an approximate molecular mass of 30 kDa, recognized by an antibody directed against the C terminus of APP (Y188) and absent in the brains of APP-knockout mice. The molecular mass of this newly identified CTF suggests an additional physiological cleavage of APP occurs amino-terminal to the established cleavage sites of β - and α -secretases, termed η -cleavage of APP (Willem et al., 2015; Tian and Koo, 2015). η -Secretase, also referred to as membrane-type 5-matrix metalloproteinase (MT5-MMP) or MMP-24 (Xu et al., 2024; Baranger et al., 2016), is a glycosylated trans-membrane protease that depends on Zn²⁺ for its activity. This enzyme is activated intracellularly by the Ca²⁺-dependent proprotein convertase furin and is predominantly expressed in neural cells but can also be found in astrocytes, microglia, and endothelial cells (Baranger et al., 2016). η -Secretase processes APP770 by cleaving it at the N579-M580 site, yielding a soluble N-terminal fragment of 95 kDa (designated sAPP95 or sAPP η) and a membrane-bound C-terminal fragment (η -CTF or C191) (Yang et al., 2024; Afram et al., 2023) (Fig. 7). MT5-MMP cleaves peptide bonds using a zinc-dependent catalytic mechanism. A zinc ion in the active site coordinates a water molecule, which is activated by a conserved glutamate residue (D204 in human MT5-MMP) to generate OH⁻ that attacks the substrate's peptide bond. This reaction breaks the bond, allowing MT5-MMP to regulate extracellular matrix

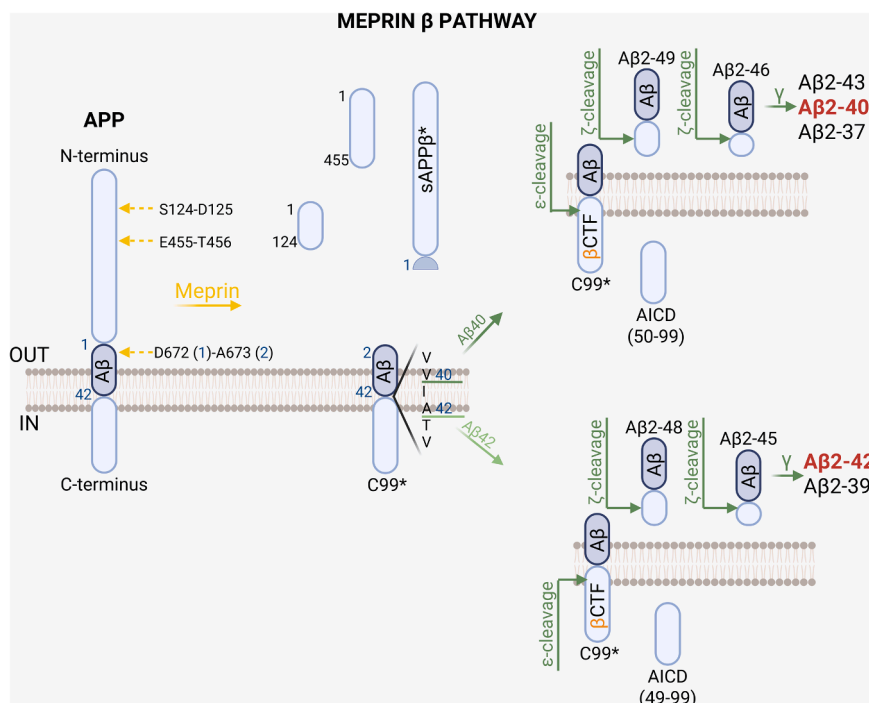


Fig. 6. Schematic representation of the amyloidogenic pathway of the APP770 processing via meprin- β . Meprin β cleaves the APP at the N-terminus of the A β sequence, generating an N-terminally truncated A β variant. This cleavage can occur independently or prior to β -secretase activity, contributing to the pool of aggregation-prone A β peptides. Created with BioRender.com.

remodeling and neural processes like synaptic plasticity (Baranger et al., 2016).

This η -CTF is further cleaved by either α -secretase or β -secretase, resulting in the release of soluble fragments A η - α or A η - β , respectively (Wu et al., 2024). The remaining C-terminal fragment undergoes cleavage by the γ -secretase complex, producing either p3 and AICD or A β and AICD, depending on the secretase activity involved (Willem

et al., 2015; Baranger et al., 2016; Dunot et al., 2023) (Fig. 7). To assess the relevance of η -secretase in AD development, studies in 5xFAD transgenic mice were conducted. In these mice, absence of MT5-MMP significantly improved pathological conditions, evidenced by decreased A β and C99 levels in the neocortex and hippocampus and reduced neuroinflammation. It also prevented deficits in long-term potentiation and spatial learning associated with the hippocampus.

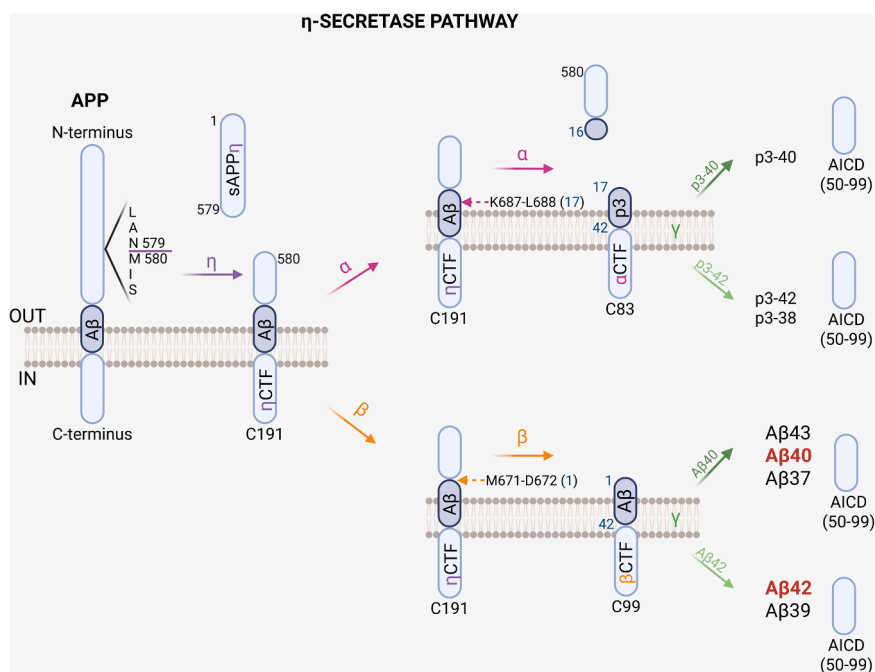


Fig. 7. Schematic representation of the amyloidogenic pathway of the APP770 processing via η -secretase. η -Secretase initiates APP cleavage by generating sAPP η and a membrane-bound η -CTF. η -CTF is further processed by α - or β -secretase into A η - α or A η - β , and the leftover fragment is cleaved by γ -secretase to produce either p3 and AICD or A β and AICD. Created with BioRender.com.

Collectively, these findings highlight MT5-MMP as an enzyme involved in APP processing that exhibits pro-amyloidogenic and synaptotoxic properties (Baranger et al., 2017; Mensch et al., 2021).

3.2.4. θ secretase pathway

BACE2, also known as memapsin 1, is a type I transmembrane aspartic protease of the pepsin family and a close homolog of BACE1, which is abundantly expressed in the brain (Ghosh et al., 2019; Sáez-Valero and Pérez-González, 2023; Wang et al., 2019). The two enzymes share a similar structural organization and ~51% amino acid sequence identity. BACE2 transcripts are present in both the central nervous system and various peripheral tissues, though neuronal expression is markedly lower than that of BACE1 (Yan et al., 2001). The BACE2 gene is located within the Down syndrome (DS) critical region on chromosome 21, suggesting a potential role in neuritic plaque formation and AD-related dementia in DS. While transcription of BACE2 is elevated in DS, protein levels do not correspondingly increase. Evidence indicates that the increased β -secretase activity in DS arises from BACE1 rather than BACE2, contributing to AD pathogenesis in DS (Sun et al., 2006).

Studies using a stable cell line co-expressing human BACE2 and APP carrying the Swedish mutation demonstrated that BACE2 cleaves APP to produce the C80 fragment, unlike the C83 fragment generated by α -secretase. N-terminal sequencing confirmed cleavage between F690-F691 and F691-A692 (Fig. 8), identifying BACE2 as a distinct θ -secretase that cleaves within the A β domain and prevents A β generation. Thus, despite an extra BACE2 gene copy in DS, BACE2 does not underlie elevated A β production associated with AD. Rather, BACE2 acts as an anti-amyloidogenic protease, and enhancing its activity may represent a therapeutic strategy for AD. Consistently, BACE1 knockout mice show no compensatory upregulation of BACE2, reinforcing that BACE1 and BACE2 have distinct biological roles (Sun et al., 2006; Abdul-Hay et al., 2012).

Other studies have shown that APP can also be processed by BACE2 at the β -site, although this activity is normally suppressed by the juxtamembrane helix (JH) of APP. When the JH is destabilized—via specific mutations or clusterin binding—BACE2 gains access to APP and

promotes β -cleavage, exacerbating AD pathology. In aged mouse brains, both BACE2 and clusterin levels increase, correlating with enhanced β -cleavage during aging. These findings suggest that BACE2 functions as a context-dependent β -secretase and may serve as a promising therapeutic target for AD, potentially avoiding side effects associated with inhibiting BACE1 (Wang et al., 2019).

3.2.5. Caspase-like proteins mediated APP cleavage

Caspases are a conserved family of cysteine-dependent endoproteases that cleave substrates at aspartic acid residues (Dehkordi et al., 2022). Caspases consist of a variable-length amino-terminal domain, followed by large (~20 kDa) and small (~10 kDa) catalytic subunits, which together form the protease domain (Van Opdenbosch and Lamkanfi, 2019; Shrestha et al., 2020). These enzymes are classified into two main types: initiators and effectors. Initiator caspases, such as caspases-2, -8, -9, and -10, contain either a caspase recruitment domain or death effector domains in their amino-terminal regions (Wu, Lo n.d.; Shalini et al., 2015), enabling their recruitment and activation within multiprotein complexes. Effector caspases, including caspases-3, -6, and -7, are activated by initiators through cleavage at specific aspartic acid residues that separate their catalytic subunits (Shi, 2002). All caspases are initially synthesized as inactive zymogens and require proteolytic activation to function in apoptosis (Donepudi and Grütter, 2002). The activation of initiator caspases triggers a cascade, activating effector caspases and setting off a chain reaction. This process is tightly regulated, typically requiring the formation of a multicomponent complex in response to apoptotic signals, ensuring precise control over the activation of these proteases (Shi, 2002). Caspase-like proteases may also influence APP processing and the development of AD. Caspases cleave the C-terminal cytosolic region after the aspartate residue at position 739, producing a membrane-bound N-terminal fragment (APP-Ncas) and a 31-residue C-terminal fragment (APP-Ccas or C31) in the cytoplasm (Park et al., 2020; Khezri and Ghasemnejad-Berenji, 2023) (Fig. 9). Park et al. demonstrated the involvement of caspases in APP fragmentation using APP695 and APP664 mutant mouse models; here, we refer to the corresponding residue in APP770 to maintain

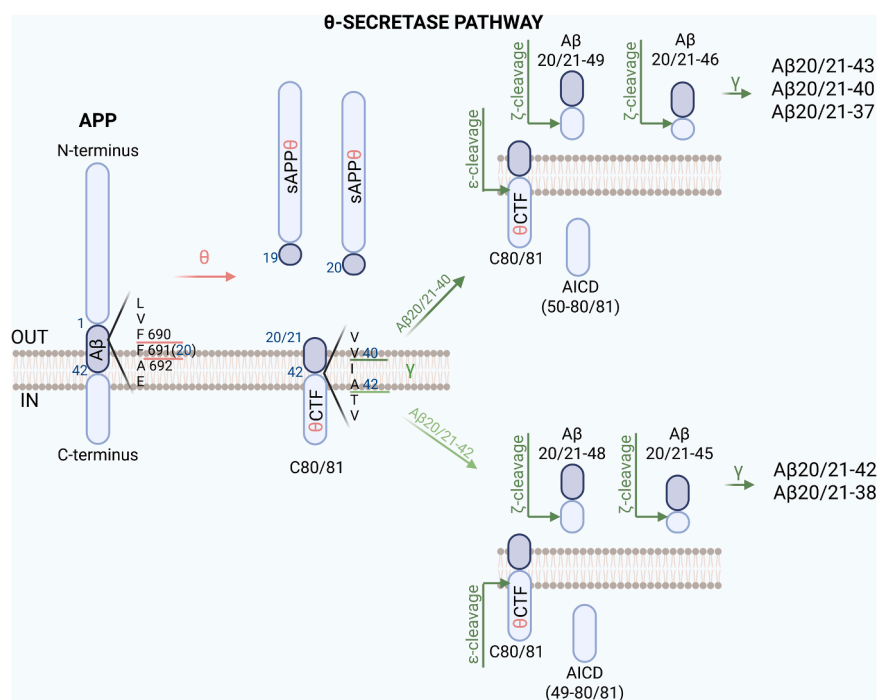


Fig. 8. Schematic representation of the amyloidogenic pathway of the APP770 processing via θ secretase. θ secretase cleaves APP predominantly within the A β region (θ -site), producing C80 that cannot generate A β after γ -secretase cleavage. This non-amyloidogenic processing contrasts with BACE1 cleavage at the β -site, which produces the amyloidogenic C99 fragment. Created with BioRender.com.

consistency throughout our analysis.

After cleavage by either α - or β -secretase 1, the resulting CTF is further processed by γ -secretase. This releases p3 or A β into the extracellular space, while the small intracellular peptide JCasp is generated. JCasp interacts with and sequesters proteins involved in the vesicle release machinery (Masi et al., 2023; Nhan et al., 2015).

Caspase 3 (EC: 3.4.22.56) appears to drive APP cleavage, and inhibiting it reduces this effect. To determine whether caspase activity directly causes increased APP695 cleavage or is merely associated with it, researchers created a caspase-insensitive APP695 D664A KI mouse with an Asp-to-Ala substitution at a critical site (Park et al., 2020). This mutation renders APP resistant to caspase cleavage, and hippocampal cultures from these mice showed resistance to A β -induced toxicity, indicating a caspase role in AD progression (Louneva et al., 2008). The active site of Caspase-3 is centered around the thiol group of C163 and the imidazole ring of H121, which plays a crucial role in stabilizing the carbonyl group of the key aspartate residue, while C163 acts as the nucleophile, attacking the peptide bond and facilitating its cleavage. Additionally, C163 and G238 contribute to the stabilization of the tetrahedral transition state of the substrate-enzyme complex through hydrogen bonding interactions (Wójcik et al., 2024).

Studies also reveal caspase 8's (EC: 3.4.22.61) role in AD, where removing caspase 8 along with receptor-interacting protein kinase-3 (RIPK3) lowers A β buildup, while removing RIPK3 alone does not (Kumar et al., 2023; Choi et al., 2023). This suggests caspases impact early AD processes, challenging the traditional view of caspases as merely involved in cell death (Khan et al., 2022). Caspases 3 and 8, in particular, seem crucial for initiating AD pathology by aiding in A β plaque formation, a key early-stage event in AD. Caspase 8 activation prompts caspase 3 activity, suggesting that caspase 8 influences A β processing indirectly by triggering caspase 3. In AD, inflammation and higher TNF α levels activate death receptors, leading to caspase 3 and 8 activation and triggering a feedback loop that sustains A β production and neuroinflammation. A β itself may interact directly with death

receptors and, through receptors for advanced glycation end-products, cross the BBB, contributing to endothelial inflammation and apoptosis. These processes likely drive the neurovascular changes observed in AD (Wójcik et al., 2024; Guo et al., 2020).

A β induces apoptosis in both endothelial cells and neurons. In neuronal cells, caspase 8 activation by A β leads to caspase 3 and 9 (EC: 3.4.22.62) activation, as shown in studies using caspase 8 knockdowns, where apoptosis rates fell significantly (Qian et al., 2015). Although caspase 8 silencing reduces A β -induced apoptosis, it does not prevent necroptosis, indicating caspase 8 primarily promotes cell death in AD rather than inhibiting necroptosis. A β may activate caspase 8 via death receptors like the Tumor necrosis factor-related apoptosis-inducing ligand (TRAIL) receptor, with A β binding observed on TRAIL receptors. Inhibiting TRAIL reduces A β -driven apoptosis, emphasizing TRAIL's importance in caspase 8 activation in AD (Fossati et al., 2012; Anzovino et al., 2023).

These findings reveal caspases' dual roles in AD—both promoting A β cleavage and mediating cell death. Caspase 8, in particular, activates downstream caspase 3, essential for AD-related A β processing (Louneva et al., 2008).

3.2.6. RHBDL4-mediated APP cleavage

Beyond the well-known AD-related enzymes, others may also significantly influence APP processing. Exploring these additional enzymes sheds light on APP metabolism and highlights potential new targets for AD treatment. Endoplasmic reticulum-resident intra-membrane rhomboid protease (RHBDL4; EC: 3.4.21.105), together with four other enzymes, belongs to rhomboid proteases involved in the mammalian secretory pathway. It contains a conserved rhomboid structure with a catalytically active serine-histidine dyad, as well as a ubiquitin-interacting motif and a VCP/p97-binding motif in its C-terminal region. Studies using model substrates have shown that RHBDL4 targets unstable proteins for cleavage, either within their trans-membrane domain or luminal segment. In RHBDL4, His204 located in

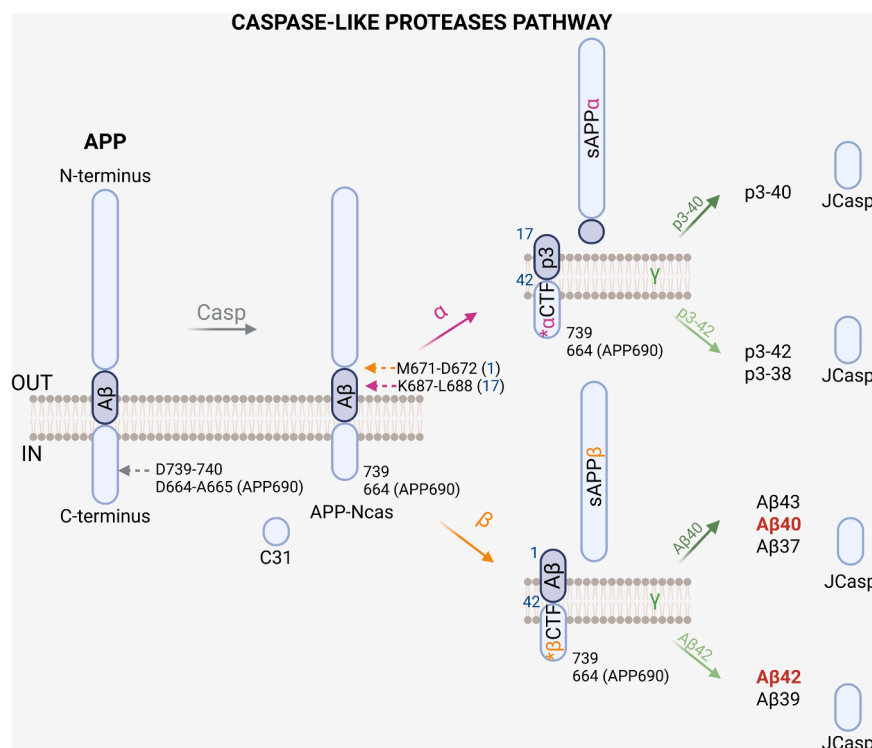


Fig. 9. Schematic representation of the amyloidogenic pathway of the APP770 processing via caspase-like proteases. Caspase-like proteases cut APP producing a membrane-bound APP-Ncas and a C31 in the cytoplasm. Subsequent cleavage by α - or β -secretase, followed by γ -secretase, releases p3 or A β outside the cell and generates the intracellular peptide JCasp. Created with BioRender.com.

transmembrane helix 6 functions as a general base by deprotonating the hydroxyl group of Ser144 in helix 4, generating a reactive serine alkoxide (Knopf et al., 2020). This activated Ser144 then performs a nucleophilic attack on the carbonyl carbon of the scissile peptide bond within the substrate's transmembrane domain. The resulting tetrahedral intermediate is stabilized by backbone amides in the oxyanion hole, which facilitates cleavage of the peptide bond. Following hydrolysis, the substrate fragments are released, and the active site is regenerated, ready to catalyze subsequent reactions.

Following RHBDL4 cleavage, protein fragments are removed in a p97-dependent process via the ERAD system. RHBDL4 can also target and cleave aggregation-prone proteins that lack a membrane anchor (Knopf et al., 2020; Bock et al., 2022; Knopf et al., 2024). Its influence on AD development is not fully understood. Studies show that APP can act as a substrate for this enzyme and be cleaved multiple times in its ectodomain. These cleavages generate 70–73 kDa N-terminal fragments and 10–25 kDa CTFs (Paschowsky et al., 2016). RHBDL4 has been shown to produce large APP CTFs that closely resemble those generated by η -secretase or MT5-MMP, which produce A η fragments. Furthermore, studies indicate that in vitro A η -like fragment generation occurs without the involvement of α -, β -, or γ -secretases. These fragments are likely formed in the ER, as they do not appear on the cell surface. Interestingly, inherited familial APP mutations do not seem to influence this processing pathway. This data emphasizes RHBDL4's role in η -secretase-like processing of APP, establishing it as a physiologically relevant protease for APP (Penalva et al., 2024; Paschowsky et al., 2018).

However, other studies showed that RHBDL4 significantly lowers the level of A β peptide (A β 38 levels were reduced by 83%, A β 40 by 69%, and A β 42 by 57%). Based on these results, it is suspected that RHBDL4-mediated APP cleavage in the ER may bypass classical amyloidogenic APP processing (Paschowsky et al., 2016).

4. Conclusion

Although A β accumulation is a defining hallmark of AD, it is also implicated in a range of other neurological conditions. In AD, the A β 42 isoform is particularly prone to aggregation, forming neurotoxic plaques that drive progressive neurodegeneration. Further, A β deposition is also observed in CAA, DS, and other disorders, including PD, PDD, DLB and Huntington's Disease. Additionally, in brain injuries, A β accumulation contributes to progressive neuronal damage. These findings underscore the broader relevance of A β pathology beyond Alzheimer's.

Targeting A β has gained attention in AD treatment (Zhang et al., 2023; Tan et al., 2014; Villain et al., 2022), with some therapies recently approved. The novel treatments include anti-A β monoclonal antibodies like Aducanumab (Aduhelm) and Lecanemab (Leqembi), which aim to clear amyloid plaques and slow cognitive decline in early AD. However, their clinical effectiveness remains debated, with modest cognitive benefits and safety concerns, such as amyloid-related imaging abnormalities, including brain swelling or/and bleeding. Another therapeutic strategy targeting A β formation that is under investigation involves the use of BACE1 inhibitors, which target β -site BACE1—the enzyme responsible for the initial step in APP processing. Although this approach is designed to reduce the production of A β peptides, it has not yet resulted in any FDA-approved treatments. Multiple clinical trials of BACE1 inhibitors have failed to achieve efficacy endpoints, largely due to limited impact on disease progression and unfavorable safety profiles. These setbacks can be attributed to several critical challenges, including insufficient specificity of the therapeutic agents, the widespread physiological roles of BACE1, disruptions to A β homeostasis, and the inability to initiate treatment during the optimal therapeutic window. Moreover, the lack of accurate and reliable translational models that faithfully replicate human disease pathology further hampers the development of effective BACE1-targeted therapies.

The significant barrier to success lies in the adverse side effects often associated with A β -targeting interventions. In many cases, the intended

therapeutic effects are offset by the oxidative stress these treatments induce, particularly in neurons—cells that are exceptionally vulnerable to oxidative damage. This creates a vicious cycle where efforts to reduce A β may simultaneously contribute to neuronal injury, further complicating disease progression. The challenge, therefore, lies in developing therapeutic strategies that do not exacerbate oxidative stress but instead adopt a multi-targeted approach: simultaneously inhibiting A β production, reducing ROS, and promoting A β clearance. A dual or even triple mechanism of action may be necessary to achieve effective and safe neuroprotection in Alzheimer's disease, ideally incorporating strategies that also target tau pathology.

Targeting the proteases involved in APP processing remains a promising strategy for modulating A β production. Nonetheless, a deeper understanding of the regulatory mechanisms that control these proteases is crucial. It is insufficient to merely inhibit their activity; a comprehensive understanding of the factors that regulate their upregulation or altered activity under pathological conditions is essential. Exploring the chemical and physiological principles that influence secretases expression, activation and the enzyme's catalytic activity rate will pave the way for more selective and safer therapeutic interventions. By intervening at the level of these upstream regulatory pathways, it may be possible to fine-tune APP processing without disrupting the broader proteolytic systems essential for normal cellular function.

In this rapidly evolving field, our goal was to present a systematic and interdisciplinary update on the current literature concerning the proteases involved in APP processing, highlighting the cleavage site and catalytic mechanism. In conclusion, this review delivers an interdisciplinary synthesis of the proteolytic networks governing APP processing aiming to deepen insights that could contribute to the development of effective treatments for diseases associated with A β accumulation.

By integrating the structural, catalytic, cellular, and sequential hierarchy of APP proteolysis, we show the logic enzymatic cascade that shapes APP fragmentation and regulates neurodegenerative cascades. This mechanistic framework clarifies critical proteolytic control nodes and provides a rational foundation for precision-targeted therapeutic strategies in diseases associated with A β accumulation such as AD.

List of abbreviations

AcD	acidic domain
ACh	acetylcholine
AD	Alzheimer's disease
ADAM10	disintegrin and metalloproteinase domain-containing protein 10
AEP	Asparagine endopeptidase
AICD	APP intracellular domain
Aph-1	Anterior pharynx defective-1
ApoE	Apolipoprotein E
APP	Amyloid β precursor protein
APP-Ncas	Membrane-bound N-terminal APP fragment
A β	Amyloid β
A β O	Amyloid β oligomer
BACE1	β -site APP-cleaving enzyme-1
BBB	blood–brain barrier
BDNF	Brain-derived neurotrophic factor
C31	31-residue C-terminal fragment
CAA	Cerebral amyloid angiopathy
CAPPD	central APP domain
C/EBP β	CCAAT/enhancer-binding protein beta
CuBD	copper-binding domain
CSF	Cerebrospinal fluid
CTF	Membrane-bound C-terminal fragment
DLB	Dementia with Lewy Bodies
DS	Down syndrome
EOAD	Early-onset AD
ER	Endoplasmic reticulum

fAD	Familial AD
GFLD	growth factor-like domain
KPI	the Kunitz protease inhibitor
LOAD	Late-onset AD
LTP	Long-term potentiation
MT5-MMP	Membrane-type 5-matrix metalloproteinase
NFTs	Neurofibrillary tangles
OH-	hydroxide ion
PD	Parkinson's Disease
PDD	Parkinson's Disease with Dementia
Pen-2	Presenilin enhancer-2
PS	Presenilin
RHBDL4	Endoplasmic reticulum-resident intramembrane rhomboid protease
RIPK3	receptor-interacting protein kinase-3
ROS	Reactive oxygen species
RS—	sulfur-containing molecule
sAPP α	Soluble APP α
sAPP β	Soluble APP β
TRAIL	Tumor necrosis factor-related apoptosis-inducing ligand
VPS34	vacuolar protein sorting 34

CRedit authorship contribution statement

ZS: Conceptualization, Investigation, Writing-Original draft, Visualization. EWT: Resources, Supervision, Conceptualization, Investigation, Writing-Reviewing and Editing. All authors who contributed to this manuscript approved the submitted version.

Declaration

Ethics approval and consent to participate.

Consent for publication

Not applicable

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Not applicable.

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